

Decentralizing Demonstration to Accelerate Learning in Agriculture*

Florence Kondylis John Loeser Mushfiq Mobarak
Maria Jones Daniel Stein

October 10, 2025

Abstract

In a field experiment, farmers learn about a new flood-saline-resilient seed from a single, centralized demonstration plot in some Bangladeshi villages, while the same demonstration kit is shared by multiple farmers in other villages. The decentralized modality allows more farmers to experience the technology for themselves or through their near-neighbors, and increases adoption one year after demonstration, especially in flood-affected villages where the benefits of adoption are concentrated and observed. The effects of decentralizing demonstration disappear two years after demonstration. Adoption and subsequent disadoption patterns are consistent with decentralized demonstration accelerating learning the returns of a customized technology.

*Kondylis: World Bank, fkondylis@worldbank.org; Loeser: World Bank, jloeser@worldbank.org; Mobarak: Yale University, NBER, ahmed.mobarak@yale.edu; Jones: World Bank, mjones5@worldbank.org; Stein: Giving Green, danstein1@gmail.com. We thank Paul Christian, Kyle Emerick, Andrew Foster, Erin Kelley, Sylvie Lambert, Arianna Legovini, Karen Macours, Jeremy Magruder, and seminar audiences at IFPRI, Naples, NEUDC, Paris-Dauphine, Paris School of Economics, Princeton University, Royal Holloway, UC Berkeley, University of Virginia, Wageningen University, and the World Bank. We thank the Global Agriculture and Food Security Program, the World Bank Bangladesh Country Management Unit, the World Bank i2i, RSB, and SAFANSI Trust Funds for generous research funding. Kieran Byrne and Marcela Mello provided fabulous research assistance; Kristoffer Bjarkefur, Cynthia Sobieski, Christina Seybolt, Md. Munstasir, Mehrab Ali, and Ahmad Wahdat managed impressive field work. We thank Innovations for Poverty Action in Bangladesh for data collection work. Finally, we thank the IAPP implementation unit staff for being outstanding research partners. The study protocols were approved by Yale University IRB (#1208010629) and IPA IRB (#00006083), and is registered in the AEA RCT Registry (AEARCTR-0010751). Computational reproducibility verified by DIME Analytics. The views expressed in this manuscript do not reflect the views of the World Bank. All errors are our own.

1 Introduction

Accelerating agricultural technology adoption in developing countries has the potential to substantially reduce global poverty (World Bank, 2007). One central challenge is the substantial heterogeneity in the returns to technologies across space and across farmers (Suri, 2011). To assess whether adoption is productive, farmers must translate information about new technologies to their local conditions (Suri and Udry, 2022). In this context, setting up *demonstration plots* is a common extension strategy to generate local information for farmers, and persuade them to adopt new technologies. While this model has shown some promise (Dar et al., 2023), it often fails to trigger large learning gains (Kondylis et al., 2017).

The optimal design of the demonstration plot and its impacts on technology adoption depend on exactly how farmers learn. Demonstration can be either centralized and conducted on a single large plot, or those same resources can be shared by a larger number of farmers, each of whom demonstrates independently on smaller plots of land. While *decentralizing* demonstration allows more farmers to gain experience first-hand with the technology, or second-hand through their near-neighbors, it comes with tradeoffs. Decentralizing demonstration reduces the scale of each demonstration, and forgoes benefits of selective targeting of farmers to lead demonstrations (BenYishay and Mobarak, 2019; Beaman et al., 2021; Akbarpour et al., 2023). These sets of countervailing forces can influence the speed and spread of learning and technology adoption (Foster and Rosenzweig, 1995).

In this paper we randomly vary the nature of demonstration across 110 villages in Bangladesh, ranging from traditional demonstration by a single farmer, shared demonstration by a few, to decentralized demonstration in which many farmers gain first-hand experience with the new technology. We then combine our experimental results with a model of learning to quantify the tradeoff between the scale and targeting benefits of centralized demonstration plots versus the benefits of experiential learning and social proximity to demonstrators.

We conduct this experiment in coastal Barisal district in southern Bangladesh, a region that has become increasingly vulnerable to salinity intrusion caused by climate change-induced storm surges and flooding. The Government of Bangladesh is promoting new, higher-yield, flood-saline-resilient seeds for traditional pulses. As storm surges and flooding are expected to become more frequent and more severe with climate change and sea level rise, the main goal of promoting the new seeds is to improve resilience to salinity intrusion. A secondary environmental goal is to encourage farmers to shift away from profitable dry-season paddy cultivation which is dependent on groundwater irrigation.¹

¹New flood-saline-resilient seeds were promoted for five crops: lentils, mung, mustard, sesame, and

These features of our context and technology foreshadow our eventual findings. These are not new crops, but new seed varieties. To decide whether to adopt, farmers must learn the local productivity of the new varieties, as opposed to how to cultivate them. Farmers adopt sufficiently productive varieties, but switching to the new seed varieties will not be profitable for every farmer. The varieties may be the most productive only in the presence of floods, and some farmers may be unproductive at cultivating pulses, especially relative to paddy.

Each village in our experiment has one farmer group, and our treatments are randomly varied at the village (or equivalently, farmer group) level. Our benchmark treatment is the “*regular*” demonstration protocol where the extension worker selects one farmer within a “farmer field group” to receive a demonstration package of seed and fertilizer sufficient to cultivate 0.4 hectares. This designated farmer is asked to run a one-season demonstration plot and share insights with others in the group.

In a second treatment called “*shared*” demonstration, the extension worker identifies up to four farmers who cultivate contiguous plots and are willing to set up a shared demonstration plot. The exact same demonstration package is divided across this group of four farmers. The shared arm is designed such that demonstration is socially decentralized, but not geographically, relative to the regular arm.

In a third set of randomly selected villages, up to twelve interested farmers share the same demonstration package amongst themselves, so that they can experiment with the new seed on their own plots. Their plots need not be contiguous, so this third “*decentralized*” arm triggers both social and geographic decentralization of demonstration.

We compare these three arms to a control arm in which the new seed is promoted by the extension agent in the villages, but no demonstration resources are provided. The promoted seeds were locally multiplied and widely available in markets across all study villages.

The program evaluation shows that demonstration activities of any type increase adoption of new seeds by 4.5 percentage points in the two years after demonstration (beyond mechanical effects during the demonstration year) relative to control villages. Only 13% of farmers cultivate the average pulse in our sample, so this represents a 35% increase in adoption of a new seed variety among those who cultivate that crop. The experiment also shows that decentralizing demonstration has a larger effect on adoption one year after demonstration. The adoption rate was 6.5 percentage points higher in villages in our decentralized arm compared to regular demonstration villages. Farmers appear to learn faster under decentralized

wheat. This was among the first efforts to promote flood-saline-resilient varieties of these crops. We use the shorthand “pulses” to describe this group because the three pulses account for over 80% of the total cultivation of these five crops.

demonstration.

Further, the technology works as intended: while floods in this coastal region adversely affect yields and profits because the water can be saline, adopting the new flood-saline resistant seed variety fully eliminates those adverse effects. Consequentially, adoption the year following demonstration is concentrated in villages that experienced a flood during demonstration.² Relatedly, the yield and profit data highlight that switching to the new seeds may not be optimal for all farmers, because not all farmers are equally flood-affected. Returns to adoption therefore vary across farmers (Suri, 2011; Patel, 2024) especially compared to cultivating lucrative dry season paddy.

Given the heterogeneous returns to the new seeds across farmers, we should not expect adoption to follow an “S-curve” towards universal adoption. Strikingly, in the second year after demonstration, adoption effects persist but become smaller in treatment villages, the effects of floods on adoption dissipate, and the decentralized and shared demo arms are no longer statistically distinguishable from traditional demonstration.

We interpret the patterns of adoption and disadoption we observe under the three treatment arms through the lens of a Bayesian model of learning about a new technology (Besley and Case, 1994; Young, 2009; Beaman et al., 2021). We model farmers with prior beliefs about the distribution of returns of the new seeds. The demonstration plots provide farmers a noisy signal of the returns, which farmers use to update beliefs and decide whether to adopt. The value of the signal depends on whether that experimentation was conducted on one’s own plot (self-learning), or on the plot of others socially connected to this farmer (social learning), or on disconnected plots. Future adoption decisions then respond to the profits farmers observe when they or their social connections experiment. We show that adoption in this model can be increasing and then decreasing in learning, as noise generated along the learning path may induce adoption mistakes. This model can therefore rationalize both the disadoption we observe in the second year, as well as the adoption “S-curve” observed in other agricultural technology adoption studies (Griliches, 1957; Foster and Rosenzweig, 1995; Beaman et al., 2021).

We present two additional results suggesting the adoption dynamics we observe are driven by adoption mistakes on the learning path. First, we show veteran pulse cultivators made fewer mistakes with the new seeds—demonstration impacts on adoption in the second year fall significantly more for novice, relative to veteran, pulse cultivators. Second, we show that demonstration significantly *decreased* profits the first year after demonstration, while profit impacts are significantly larger (and positive) in the second year. Taken together,

²Cai et al. (2020) document a similar effect in China with weather insurance, when insurance is well understood.

these results suggest that farmers become more positively selected into the new seeds in the second year and fewer mistakes are made.³

We use the model to estimate the relative strength of “learning from self” and “learning from others.” We find that the effect of one’s own experimentation is seven times stronger on future adoption than the experiments conducted by a social connection. Since each farmer has about nine social connections on average, our estimates imply that roughly half of all knowledge diffusion that gets generated by our demonstration experiments can be attributed to *social* learning while the other half is “*self*” learning.

Through the lens of this learning model, the reason why decentralized demonstration produces the largest treatment effects on adoption is that the largest number of farmers get to participate in demonstration and directly experience the new technology both for themselves and through their near-neighbors. We present empirical evidence that the smaller scale of experimentation under decentralized demonstration, and moving away from a carefully selected single demonstrator, did not undermine the value of the information generated. Farmers learned from themselves and from others in their village, and decentralizing demonstration accelerated learning in the presence of social frictions.

We describe the study context in Section 2, and our experimental design and data in Section 3. Section 4 presents our experimental estimates of the impacts of demonstration. Section 5 presents the model of learning. Section 6 tests the model and quantifies the learning mechanisms. Section 7 concludes.

2 Context

2.1 Setting

The Ministry of Agriculture of Bangladesh launched the Integrated Agricultural Productivity Project (IAPP) in 2011, the first large-scale effort to promote flood-saline-resistant seeds. The project spanned across eight districts from August 2011 to December 2016 under the supervision of the World Bank. However, our study area in this paper is limited to the Barisal district.

As part of the IAPP, the Bangladesh Agriculture Research Institute (BARI) engineered

³While decentralizing speeds up the learning process, many farmers make adoption mistakes along the learning path. This may reduce welfare if the cost of those mistakes are larger than the social gains from faster learning. In a previous version of this paper (Kondylis et al., 2023), we structurally estimated the model to quantify these gains and losses. We found that, for the seeds we study, farmers’ profits were initially decreasing in learning. However, in expectation with respect to farmers’ prior beliefs over the returns of the new seeds, the positive learning externalities generated by decentralized demonstration exceed demonstration costs. Decentralizing demonstration improves welfare, on net.

new seed varieties with higher productive resilience. Extension services supported the propagation of the new seeds through Union-level seed multipliers, ensuring that farmers all across our study area could access them on local markets; as noted in [Patel \(2024\)](#), we do not find any economically meaningful price difference between traditional seeds and the flood-saline resilient varieties promoted in our study.⁴

The IAPP promoted adoption of new seeds for five traditional crops (“pulses”):⁵ lentils, mung, mustard, sesame, and wheat. Sub-Assistant Agricultural Officers (extension agents, hereafter) promoted the seeds across villages. One of the primary activities of extension agents was to form one farmer field group of approximately 25 farmers in each project village. These farmers were self-designated during a village meeting with the extension agent; the only requirement was to show interest in learning about new agricultural technologies and interest in attending monthly group meetings during the dry season. The groups discussed a range of agricultural practices and technologies, including the properties of the new seeds.

To limit the potential for our demonstration interventions to displace extension agents’ attention away from farmers in control and non-experimental villages, our experimental sample consists of 2 randomly selected villages in 55 Unions of the district (see Appendix Figure A1). Our study sample is entirely made up of farmer field group members (as there is one group in each village, farmer group analysis is equivalent to village analysis).

2.2 Dry season agriculture in Barisal

Agriculture comprises two main growing seasons in Barisal: the monsoon season (*kharif* season) spans roughly from April to October, while the dry season (*rabi* season) runs roughly from November to March. Rice is the traditional staple crop in Bangladesh, and is well-suited to cultivation in the kharif season. While productive, rice cultivation in the rabi season (*boro* rice) requires heavy irrigation which is primarily provided by groundwater.⁶ Heavy reliance on groundwater irrigation poses economic and environmental challenges: at the time of this study, heavy energy subsidies were in place to ensure the profitability of boro rice cultivation, and associated groundwater extraction was drawing down the water table.

We present basic descriptive statistics on dry season production of pulses and rice in Barisal in Appendix Table A2; our household survey data used to construct this table is

⁴The seeds promoted through the IAPP are open-pollinated varieties, so in theory farmers could recycle seeds (i.e., reproduce the improved seed from their harvest). In practice, we do not observe a meaningful degree of recycling of the improved seeds. Instead, the fact that these are open-pollinated varieties may have facilitated Union-level seed multiplication and in turn supply on local markets.

⁵Only three of these crops are pulses—lentils, mung, and mustard— but these three are 83% of post-demonstration farmer-crop-wave observations. For concreteness, we refer to all five as “pulses”.

⁶Most irrigation in Barisal originates from pump-operated shallow tube wells (67% in our study sample, compared to 5% using surface water).

described in Section 3.3.

2.3 New seeds, same old crops

The frequency of flood-saline events is increasing in the Barisal district as a consequence of climate change (Dasgupta et al., 2014). This is not unique to our study context: about 1.5 billion people are affected by floods, and a majority of them are in areas where agriculture is the main source of livelihood (Dar et al., 2013; Lane, 2024). In Barisal, tidal floods reduce agricultural productivity not only by visibly damaging crops, but also by facilitating salinity intrusion into the water table. These effects are strongest during the dry season, when the absence of rains prevents water table recharge and salinity from leaving the superficial layers of the soil.

In this context of rising seas and falling water tables, the adoption of flood-saline-resilient seeds holds potentially stark welfare implications. The improved varieties we study were developed not only with the goal of increasing the productive resilience of pulses, but also to cause farmers to move away from cultivating boro rice, reducing the use of dry season irrigation and slowing groundwater depletion.⁷

We expect farmers to predominantly learn about the productivity of the improved seeds, rather than about the shape of or inputs to the production technology.⁸ Farmers in Barisal have been well exposed to the cultivation of the pulses, and the new seeds do not affect the recommended cultivation practices for these crops.⁹ While farmers are familiar with both flood-saline-resilient and high-yielding varieties of seeds in the case of rice, the increased yields associated with a given high-yielding variety of seeds will vary markedly both across varieties and contexts—for any given variety newly introduced to a context, farmers must learn just how impactful the variety is on yields (Laaaj et al., 2020).

⁷Pulses were largely bypassed by the Green Revolution in Barisal; the delayed and relatively less effective development of improved varieties of these crops, not typically cultivated in richer countries, has been an important determinant of slow agricultural productivity growth in lower income countries (Gollin et al., 2021).

⁸While we expect farmers’ learning to focus on productivity, we do not rule out multiple dimensions of farmer uncertainty in our context. For instance, Patel (2024) documents that farmers in neighboring Khulna are uncertain about soil salinity on their plots, and increase their demand for saline-resilient varieties of paddy in response to revising their beliefs about their plots’ soil salinity upwards, despite high familiarity with saline-resilient paddy.

⁹This does not imply farmers’ input allocations will not adjust under the improved seeds: as the new seeds may both increase yields and reduce the downside risk of cultivating these crops, we may expect a “factor-deepening” effect as reported by Emerick et al. (2016).

3 Experimentally decentralizing demonstration

Demonstration resources increase both learning from self and from others: the farmer who leads the demonstration acquires first-hand experience with the technology, while other farmers in the village observe and learn from them. When demonstration resources are fixed, the optimal allocation of demonstration resources is determined by the level of demonstration decentralization that maximizes learning. Our experimental design and conceptual framework aim to identify the optimal level of decentralization and the mechanisms at play.

3.1 Experimental design

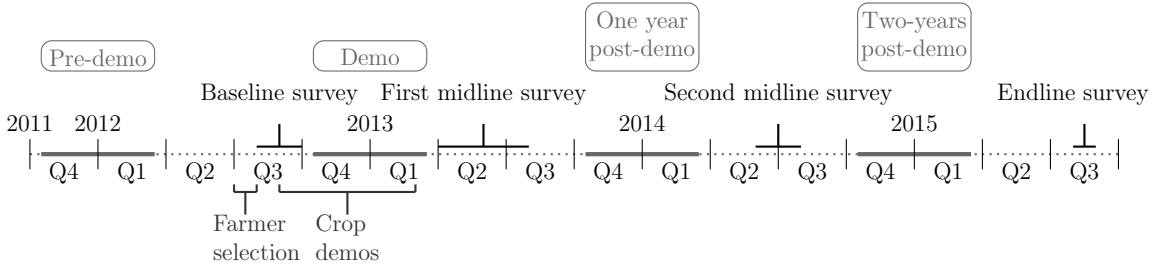


Figure 1: Timeline

Notes: Gray solid lines indicate the boro season each year, denoted *Pre-demo*, *Demo*, *One year post-demo* and *Two years post-demo*, respectively. Survey periods are highlighted in black. Dark gray brackets denote project interventions.

Timeline and experimental design We implement our intervention through the farmer field groups that the IAPP formed in each village. We present the timeline of activities associated with the IAPP and our experiment in Figure 1. Initial farmer group meetings were carried out in July 2012, a few months prior to the dry season when pulses are particularly suited for cultivation. We use these meetings to generate each village’s sampling frame for our baseline survey (see Section 3.3 for more details).

After the initial farmer group meetings, we randomly assigned villages in our study sample to one of four groups: 1) decentralized demo, 2) shared demo, 3) regular demo, and 4) a control group. Figure 2 summarizes the experimental design and sample sizes of each arm.

Control No demonstration resources were provided through farmer groups. However, extension agents participate in regular meetings with the farmers groups to discuss and promote the new seeds and other agricultural practices. In addition, the new seeds were locally multiplied at the Union-level and were available on all markets in the study area.

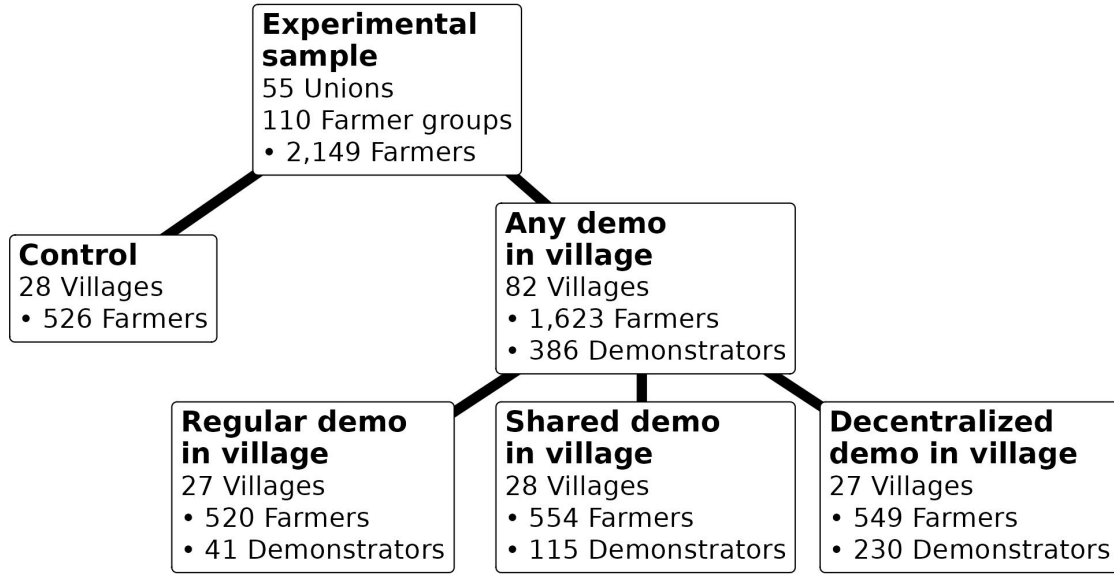


Figure 2: Experimental design

Notes: The assignment of villages across experimental arms is described in this figure. The number of farmers in our survey sample and, in villages assigned to demonstration, the number of demonstrators, associated with each experimental arm are presented in this figure.

Any demonstration After the initial farmer group meetings and treatment assignment, in August and September 2012, extension agents organized a farmer group meeting to introduce farmer-led demonstration of the new seeds in each treatment village. The demonstration was designed to complement the promotion of the new seeds by agricultural extension agents. The process of introducing demonstration of the new seeds was as follows.

First, extension agents selected up to three pulses with new seed varieties suitable for cultivation in their Union during the November 2012–March 2013 dry season (“Demo year”). This allows us to observe a demonstration crop for all villages in our study, even those not assigned to the demonstration treatment. The project then delivered one demonstration package for each selected crop per treated village and selected demonstration farmers. Each package contained sufficient quantities of fertilizer and new seeds to cultivate roughly 0.4 hectares.¹⁰

Second, during the meeting to introduce farmer-led demonstration, the extension agent

¹⁰In addition to the pulses, extension agents could also select improved varieties of dry season rice for demonstration. In contrast to the pulses, the development and promotion of improved varieties of rice is common in Barisal, and we found over 80 varieties available. We did not analyze these demonstrations, as the rice varieties available for demonstration by the IAPP were already familiar to farmers, so there was no scope for learning. Besides, rice demonstrations were unlikely to contribute to the IAPP’s objective of increasing cultivation of pulses to enable farmers to substitute away from dry season rice cultivation.

identified farmers willing to set up demonstration plots using the demonstration packages. Across treatment arms, each farmer could participate in demonstration of at most one crop.¹¹ All treatment arms have the same demonstration resources allocated across farmers, but they varied in how each package was divided across the farmers.

Regular demonstration treatment arm The extension agent designated a single farmer to receive the full package to maintain a demonstration plot on their land. This identifies the impact of the benchmark demonstration model (Kondylis et al., 2017; Beaman et al., 2021). The purpose of this model is to help other farmers learn from the demonstration, while the demonstration farmer learns by demonstrating. Farmers selected under regular demonstration may differ from other farmers; for instance, extension agents may select more central farmers, who may better facilitate the learning of other farmers, or farmers with larger landholdings, who have a sufficiently large plot to cultivate the full demonstration package.

Shared demonstration treatment arm Demonstration is socially but not geographically decentralized. For each demonstration package, the extension worker identified up to four farmers with contiguous plots to set up a shared demonstration plot. The shared arm limits the geographic spread of the demonstration while increasing the number of farmers who demonstrate relative to the regular arm. As a consequence, the scale of the demonstration and farmers’ geographic proximity to it should be similar across the shared and regular arms.

Decentralized demonstration treatment arm Demonstration is socially and geographically decentralized. The extension worker allowed up to twelve volunteer farmers to split the demonstration package and experiment with the new seed on their own plots. In this arm, the resulting demonstration “minikits” remain sufficient to cultivate at least 0.03 hectares with the new seeds, which corresponds to a fairly typical plot size in our context. This approach is in the spirit of a temporary subsidy (albeit one implemented by a quota floor) for experimenting with a new technology, as studied in Carter et al. (2021). Compared to the shared and regular arms, the decentralized arm allows more farmers to learn from their own demonstration, although the scale of each individual demonstration is smaller. In turn, farmers are more likely to be socially connected to a demonstrator and to be geographically close to a demonstration plot, so a larger number of farmers may learn from observing these smaller demonstration plots.

¹¹Out of 386 demonstrators, we observe two who demonstrated multiple crops.

3.2 Conceptual Framework

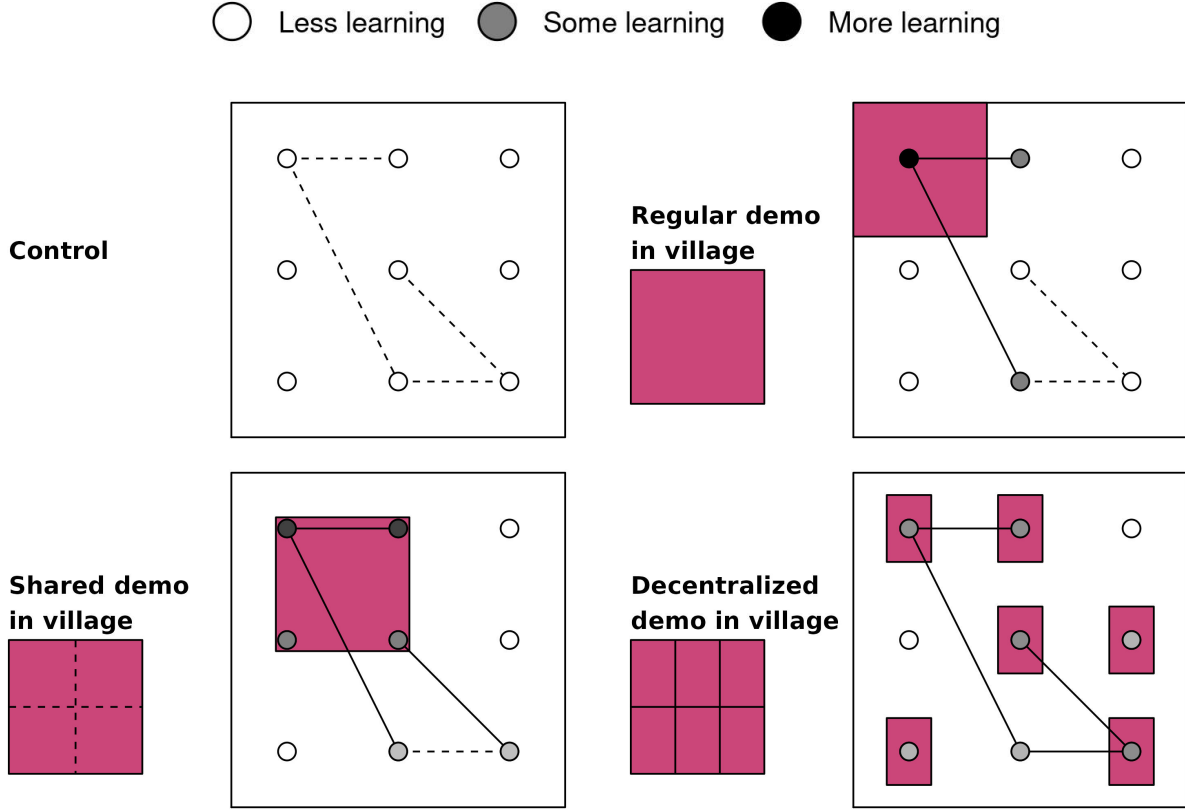


Figure 3: Linking Conceptual Framework and Experiment

Notes: A stylized representation of demonstration modalities across experimental arms, and associated learning predicted by our conceptual framework, is presented in this figure. Circles represent households. Solid and dotted lines between households represent social connections along which learning does and does not occur, respectively. Circle color represents strength of learning from demonstration. Pink rectangles represent demonstration plots, with area proportional to allocated demonstration resources.

We now discuss the mechanisms through which demonstration modalities may interact with the way farmers learn about a new technology. We develop a framework with fixed demonstration resources within a village, where the scale of each individual farmer’s demonstration is inversely proportionally adjusted to the number of farmers sharing the demonstration resources.¹²

Figure 3 illustrates our conceptual framework. Under fixed demonstration resources,

¹²While this conceptual framework discusses learning, we note that adoption of new technologies is commonly conceptualized as monotonic in learning, overcoming the empirical constraint that learning is often challenging to observe: it is assumed that increases in adoption correspond to increases in learning, and as more farmers learn, or as farmers learn more, about the technology, they become (perhaps weakly) more likely to adopt it (Griliches, 1957; Foster and Rosenzweig, 1995; BenYishay and Mobarak, 2019).

decentralization may increase or decrease learning through four different mechanisms.

First, decentralization increases the number of farmers that participate in the demonstration, increasing the opportunities for farmers to learn from their own experience.

Second, as more farmers experience the technology, there are more opportunities to learn from others. If there are “threshold effects” in social learning, a farmer may only decide to adopt once they have observed a minimum number of signals (Beaman et al., 2021). Conversely, above a certain number of signals, returns to decentralization may decrease (Foster and Rosenzweig, 1995; Conley and Udry, 2010). Depending on the relative strength of learning from self and others, decentralization may then only increase learning up to a point.

Third, learning may increase as agricultural and geographical proximity increase. If farmers learn the most from observing the technology in similar agricultural growing conditions or close to their own plot (Conley and Udry, 2010), then geographically decentralizing the demonstration will also unambiguously increase learning.

Finally, however, learning may decrease with the scale of the demonstration. Under a fixed allocation of resources, decentralization mechanically reduces the demonstration scale that farmers carry out and observe. If the scale plays an important role in determining the precision of the signal a farmer extracts from (their own and others’) experimentation (Foster and Rosenzweig, 1995), then this mechanism will interact with the relative strength of learning from self and from others. Alternatively, traditional demonstration farmers may be more central than demonstration farmers under decentralized modalities, and decentralizing may reduce learning.

There is an ambiguous effect of decentralizing demonstration activities on learning. When learning from scale is first order, the largest impacts are from providing all seed to one farmer, unless learning from self is very large relative to learning from others. When learning from decentralization is first order, the largest impacts are from providing seeds to many farmers.

By varying the degree of social and geographic decentralization of demonstration activities, our experimental design will allow us to gauge the relative importance of these mechanisms. However, as decentralization and scale interact with learning from self and others, these forces cannot be experimentally isolated. In Section 5.3, we impose restrictions on our model of learning and generate model tests to isolate the importance of decentralization and of scale.

3.3 Data

We collected four rounds of detailed household surveys, complemented by monitoring data on demonstration plots, to support our analysis; we now discuss these data and their application to our analysis. Appendix A provides a detailed description of all variables used in our analysis and details their construction, and Appendix Table A3 shows how our sample size changes across survey waves and all sample restrictions used in our analysis.

Household surveys Between 2012 and 2015, we administered four rounds of household surveys of farmer group members across our experimental sample. Each survey round was timed to immediately follow the harvest of the dry season; Figure 1 provides a detailed timeline of these survey rounds. Each survey round collected detailed records of agricultural production and technology adoption during the dry season, including adoption of the new flood-saline-resilient seeds promoted by the project. We leverage data from these surveys at the level of the village, household, household-survey wave, and household-crop-survey wave.

Baseline survey The baseline survey was administered on a sample of 1,407 IAPP farmer group members within 78 randomly selected villages of the 110 villages in our experimental sample.¹³ In addition to household characteristics, and the dry season agricultural data collected in each survey wave, the baseline survey included a module on social network linkages, in which we asked each household questions about their interactions with all other farmers in their farmer group (sampled or not). In our analysis, we consider a household to be in another household’s social network if they report speaking to them, or were reported being spoken to by them, about farming in the last three months; to ensure consistent measurement across households, for all regressions leveraging social network relationships we restrict our sample to farmers and villages surveyed at baseline.

Follow-up surveys Our next three rounds of follow-up surveys preserved and expanded the baseline sample to add 496 farmer group members across the remaining 32 villages in our experimental sample, and an additional 246 farmer group members across the original 78 villages. Regressions that leverage baseline information such as social network relationships will not include these additional 32 villages and 742 group members; in Appendix Tables A15 and A16 we show that including or dropping these villages does not meaningfully affect our main results beyond the precision of our estimates.

¹³The baseline survey covered 279 of the 386 eventual demonstrators, including 24 of 41 regular demonstrators, 81 of 115 shared demonstrators, and 174 of 230 decentralized demonstrators.

In our first follow-up survey, just at the end of the demo year, we surveyed a much smaller sample of households and oversampled households in treatment arms with more demonstrators. We use the demo year household survey in two tables and one figure. In Table 5, we include demo year data for additional power when estimating the impacts of the new seeds on yields; these results are robust to excluding demo year data. In Appendix Table A1, we use this survey to construct measures of demonstration year profits for demonstrators. In Figure 4a, we show the dynamics of adoption of improved seeds across our four years of data.

In our second follow-up survey, one year post-demonstration, we additionally asked households about their knowledge of each of the new varieties of pulses, including characteristics of the new varieties. We construct an indicator that the household reported one of the new varieties for a given crop is a high-yielding variety; building on our conceptual framework, we interpret this as a binary proxy for households’ crop-specific beliefs about the returns to the new seeds, and we incorporate this measure of beliefs into analysis of mechanisms in Appendix D.

Attrition Our main analysis focuses on impacts of demonstration one and two years post-demonstration. We attempted to revisit our full baseline sample in each of these waves; we reached between 81% and 89% of households across treatment arms during these two survey waves, and find no evidence of differential attrition across treatment arms (Appendix Tables A4 and A5).

Demonstration monitoring In addition to the household survey, we gathered operational data during the demonstration year (November 2012 – March 2013) through detailed communication with each of the extension agents. For each village, we identified the full set of crops for which demonstration packages were distributed, and which farmers demonstrated each crop. These administrative data align remarkably well with our self-reported household survey data; 5% and 15% of non-demonstrators (defined at the household-crop level) report adoption of improved seed and cultivation, respectively, for promoted crops during the demonstration year, in contrast to 80% and 94% of demonstrators.

Restriction to promoted crops Our main analysis leverages farmer-crop-survey data at the wave level and restricts to what we refer to as “promoted crops.” A key choice made by extension agents for treatment villages is the selection of pulses for demonstration of improved seeds; as we do not expect impacts of demonstration on non-demonstrated pulses, pooling demonstrated and non-demonstrated crops together in analysis would meaningfully

reduce power. However, no crops were demonstrated in control villages. We therefore identify comparison crops in control villages based on the observation that extension agents were instructed to select crops for demonstration packages based on suitability in their Union (Section 3.1); specifically, we select the crops that were demonstrated in the other sampled village in their Union (and therefore selected by the same extension agent) as comparison crops.¹⁴ We refer to comparison crops in control villages, and demonstrated crops in treatment villages, as “promoted crops”, and non-promoted pulses as “placebo crops.”

The restriction to promoted crops would result in bias if promoted crops were endogenous to treatment assignment; for instance, extension agents could have been aware of the village’s treatment status (regular, shared, or decentralized) before selecting the demonstration package, and this information could have informed their choice of demonstrated crops.

We consider four tests for this bias, which we discuss and implement in Appendix B. We compare, across treatment arms, the choice of promoted crops, baseline outcomes, and outcomes on placebo crops, and we estimate impacts controlling for crop-survey wave fixed effects. The results of these tests lead us to conclude that potential endogeneity of promoted crops to treatment assignment does not meaningfully bias our results.

3.4 Implementation fidelity

We leverage data from the monitoring survey to establish that our experimental design successfully generated different levels of social and geographic decentralization of demonstration activities across arms as Figure 3 illustrates. We present descriptive statistics on the implementation of our experimental design in Table 1. As discussed in Section 3.3, promoted crops are balanced across control and treatment villages and range in number from 1.61 to 1.88 crops. However, no demonstration farmers were designated in control villages, and no demonstration plots were established.

In villages assigned to the regular treatment arm, adherence to treatment was high with on average 0.82 demonstration plots being set up for each promoted crop. Each farmer in a regular village had a 4.2% chance of demonstrating each promoted crop (roughly one farmer per 25 farmer group members), and had on average 0.46 connections to a demonstration farmer for a given promoted crop.

In villages assigned to the shared treatment arm, implementation fidelity was also high with on average 2.58 demonstration farmers setting up just 1.11 demonstration plots for each promoted crop. The social decentralization of demonstration activities in the shared arm resulted in the average farmer group member in a shared village having a 12.9% chance of

¹⁴In 2 Unions, both villages were randomly selected to be control villages; these 4 villages are excluded from all analysis, as comparison crops are not defined.

Table 1: Demonstration descriptive statistics

	Demo in village			
	Control	Regular	Shared	Decentralized
	(1)	(2)	(3)	(4)
Village				
# of promoted crops	1.88	1.85	1.61	1.85
# of demo farmers	0.00	1.52	4.11	8.52
Village-crop				
# of demos	0.00	0.82	2.58	4.62
# of demo plots	0.00	0.82	1.11	4.62
Farmer-crop				
Demo	0.000	0.042	0.129	0.223
# of connections to demo	0.000	0.463	0.952	2.683

Notes: Sample averages of implementation variables by treatment arm across village, village-crop, and farmer-crop observations are presented in this table.

demonstrating and having 0.95 connections to a demonstration farmer, for a given promoted crop.

In villages assigned to the decentralized treatment arm, the level of decentralization of demonstration activities was successfully higher than in the other treatment arms, with 4.62 demonstration farmers each setting up a demonstration plot for each promoted crop. Consequently, for each promoted crop, the average farmer group member in decentralized villages had a 22.3% chance of demonstrating and had on average 2.68 connections to a demonstrator.

4 Experimental results

We leverage the experimental variation produced by our RCT to produce three key results. First, we show that assigning demonstration resources to villages increases farmer group members' adoption of the promoted improved seed varieties (Figure 4a; Section 4.1). Second, the resulting adoption fully offsets the negative impact of flood-saline events over the course of our study, as the improved seed increases crop resilience to saline flooding (Figure 4b; Section 4.2). Third, pooled estimates mask important dynamics (Section 4.3). One year post demonstration, the social and geographical decentralization of demonstration to many farmers additionally increases technology adoption in the village (Figure 4c; Section 4.4).

In the long run, the impacts of demonstration on adoption persist but decrease (Figure 4a), while the additional impacts of decentralizing demonstration on adoption vanish (Figure 4c).

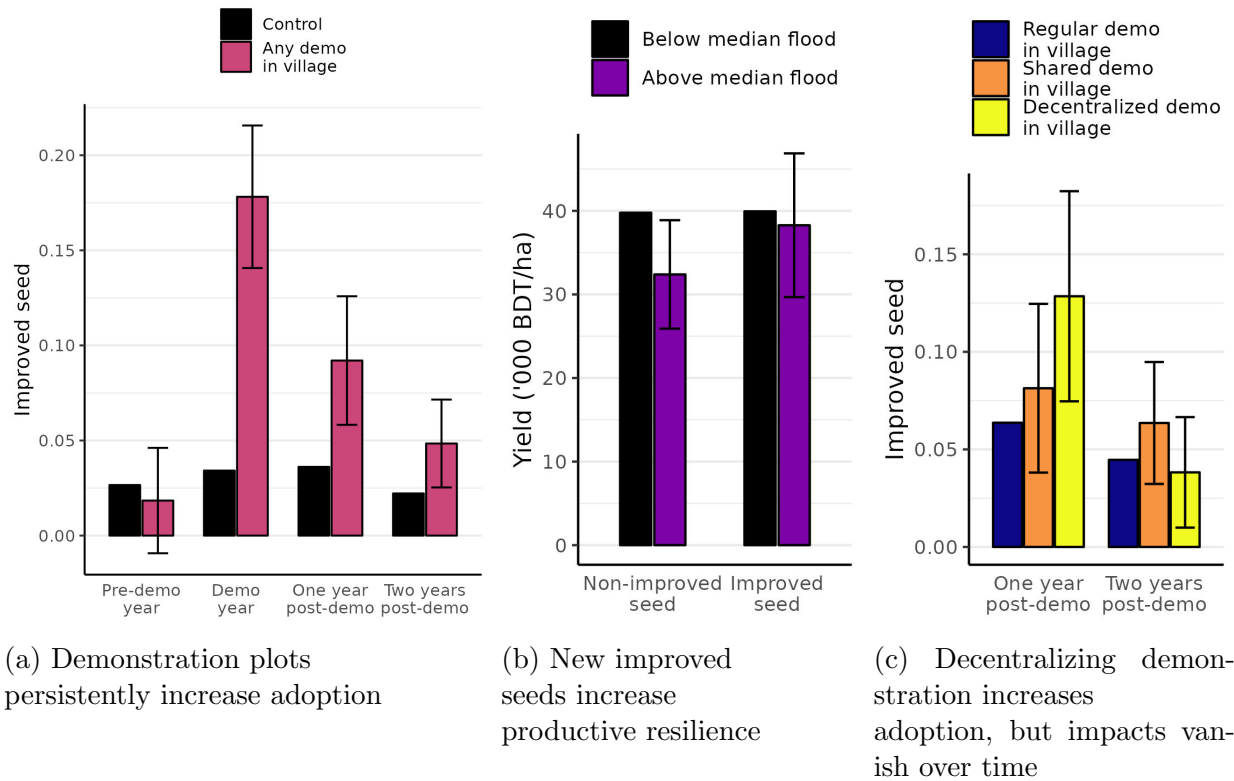


Figure 4: Demonstration and decentralizing demonstration increase adoption of a productive technology

Notes: In Panel (a), control mean outcomes are presented in black, and treatment mean outcomes are presented in pink with 95% confidence intervals on treatment effects. Survey waves are on the horizontal axis. In Panel (b), non-improved seed, below median flood, mean outcomes, plus estimated differences from Equation 3, are presented. 95% confidence intervals are presented on effects of above median flood conditional on improved seed adoption. Improved seed adoption is on the horizontal axis. In Panel (c), regular demo in village mean outcomes are presented in blue, and shared demo in village and decentralized demo in village mean outcomes are presented in orange and yellow, respectively, with 95% confidence intervals on effects relative to regular demo in village. Survey waves are on the horizontal axis.

4.1 Demonstration plots increase adoption

Empirical strategy We start by estimating farmer-crop level impacts of assigning demonstration resources to a village, comparing outcomes in control villages and demonstration villages in the two dry seasons that follow the demonstration season.

As discussed in Section 3.3, we focus our analysis throughout the paper on promoted crops; we do so for power, as we do not expect effects of demonstration on non-promoted

crops. In Appendix B, we present evidence that this restriction does not bias our results.

To estimate these impacts of assignment to demonstration on villages, we estimate:

$$Y_{hcv t} = \alpha_t + \beta_t \text{Any demo in village}_v + \epsilon_{hcv t} \quad (1)$$

where $Y_{hcv t}$ is outcome Y , for household h and promoted crop c , located in village v , in survey wave t . $\text{Any demo in village}_v$ is an indicator for village v being assigned demonstration resources under any treatment arm (regular, shared, or decentralized), and α_t are survey wave fixed effects.

We impute zeros for outcome variables, including adoption of improved seeds and profits, when the household does not cultivate the crop, with the exception of yields and fertilizer use. We therefore interpret estimated impacts on yields and fertilizer use as a combination of direct and selection effects.

Balance and attrition We establish that our village-level randomization achieved balance for key baseline farmer characteristics, both individually and jointly, across demonstration and control arms (Table 2). In Appendix Table A23, we show that inference in our tests of balance is not affected when we implement randomization inference, and we therefore instead present robust standard errors clustered at the village and associated p-values throughout our analysis.¹⁵ In Appendix Table A4, we establish that attrition was low (12.5% to 18.9%) one and two years post-demonstration and not significantly affected by assignment to demonstration.

Results Pooling across the two years following the initial demonstration year, we find that assigning demonstration resources increases adoption of the new seed among farmer group members by 4.5 percentage points, or a 145% increase over the control mean (Column 1 of Table 3).

Consistent with the objective of enabling substitution away from dry season paddy and towards pulses through the promotion of the new seeds, we find suggestive (but statistically insignificant) evidence of increased cultivation of promoted crops (Column 2 of Table 3) and decreased cultivation of paddy (Column 6 of Table 3). While neither of these coefficients are significant, we consider them further when we estimate heterogeneous effects with respect to experience with promoted crops.

We find no average impacts of being assigned to demonstration on yields or profit for the

¹⁵We present estimates of Equation 1 with randomization inference p-values included in Appendix Table A25. Inference is in general unaffected, although our estimates of the impacts of demonstration on fertilizer use fall marginally below conventional thresholds of statistical significance.

Table 2: Balance across control and treatment and across treatment arms

	Mean (SD)				Coef. (SE) [p]		
	Any demo in village				Difference		
	Control	Regular demo in village	Shared demo in village	Decentral- ized demo in village			
		(1)	(2)	(3)	(4)	(2, 3, & 4) - (1)	(3) - (2)
Age of HHH (Years)	48.69 (11.93)	48.71 (11.91)	48.32 (11.85)	48.53 (12.56)	-0.18 (0.83) [0.832]	-0.39 (0.97) [0.688]	-0.18 (1.14) [0.872]
Any pulses cultivation	0.203 (0.403)	0.240 (0.428)	0.246 (0.431)	0.160 (0.367)	0.012 (0.064) [0.850]	0.006 (0.068) [0.934]	-0.080 (0.056) [0.153]
Veteran cultivator (Any promoted crop cultivation)	0.117 (0.323)	0.100 (0.300)	0.126 (0.332)	0.117 (0.322)	-0.003 (0.061) [0.965]	0.026 (0.046) [0.573]	0.018 (0.041) [0.667]
Any pulses improved seed	0.050 (0.218)	0.076 (0.266)	0.052 (0.223)	0.040 (0.196)	0.006 (0.027) [0.834]	-0.024 (0.028) [0.392]	-0.036 (0.027) [0.181]
Any irrigation	0.463 (0.499)	0.367 (0.483)	0.406 (0.492)	0.419 (0.494)	-0.065 (0.111) [0.561]	0.039 (0.118) [0.740]	0.052 (0.119) [0.661]
Total plot area (ha)	0.202 (0.300)	0.163 (0.222)	0.198 (0.246)	0.171 (0.354)	-0.024 (0.039) [0.537]	0.035 (0.039) [0.369]	0.008 (0.050) [0.873]
# of connections	9.05 (6.65)	10.75 (6.43)	7.19 (5.04)	10.50 (5.76)	0.37 (1.62) [0.817]	-3.56 (1.57) [0.024]	-0.25 (1.75) [0.888]
# of observations (Farmer)	281	341	382	375			
# of clusters (Village)	15	19	21	21			
Omnibus F-stat [p]					0.15 [0.994]	1.33 [0.263]	0.96 [0.470]

Notes: Columns 1, 2, 3, and 4 present the baseline mean of the dependent variable and the baseline standard deviation of the dependent variable in parentheses, for farmers in control and regular, shared, and decentralized demo villages, respectively, and the total number of farmer and village observations. Columns 5, 6, and 7 present differences, with robust standard errors clustered at the village level in parentheses, and p-values in brackets.

promoted crops (Columns 3 and 5, Table 3). We revisit this finding in Section 4.2 (with the impacts of flood-saline shocks) and in Section 5.2.2 (with dynamic impacts).

Interestingly, we find a positive and significant impact on fertilizer expenditures (Column 4, Table 3)—we interpret this result as evidence of capital deepening in response to reduced downside risk with the flood-saline resilient seed, as in Emerick et al. (2016). This effect is economically meaningful, and corresponds to a 39% increase over the control mean.

Table 3: Demonstration increases adoption of improved seed

	Promoted crops, One and two years post-demo					
	Improved seed	Cultivation	Yield ('000 BDT/ha)	Fertilizer ('000 BDT/ha)	Profit ('000 BDT/ha)	Paddy cultivation
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.045 (0.013) [0.000]	0.047 (0.034) [0.163]	-3.401 (3.344) [0.310]	2.019 (0.884) [0.023]	-0.572 (0.681) [0.401]	-0.111 (0.095) [0.245]
Control mean	0.031	0.130	42.746	5.149	1.874	0.438
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	5,533	5,533	912	912	5,533	2,818
# of clusters (Village)	96	96	81	81	96	96

Notes: Selected regression coefficients from Equation 1 are presented above, with included control variables listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

Who adopts? One key determinant of the decision to adopt the new seeds is the joint decision to cultivate the associated promoted crop. As just 12% of farmers cultivated any of their village’s promoted crops at baseline, we investigate baseline cultivation of promoted crops as a dimension of heterogeneity. This heterogeneity is particularly relevant as it captures baseline differences in determinants of crop choice, such as market access, experience, and plot characteristics. We refer to farmers who did and did not cultivate promoted crops at baseline as veteran and novice cultivators, respectively, following [Conley and Udry \(2010\)](#). We modify Equation 1 to allow for heterogeneity with respect to veteran status, and estimate:

$$Y_{hcv} = \alpha_t + \beta_1 \text{Veteran cultivator}_{hv0} + \beta_2 \text{Any demo in village}_v + \beta_3 \text{Veteran cultivator}_{hv0} \times \text{Any demo in village}_v + \epsilon_{hcv} \quad (2)$$

where $\text{Veteran cultivator}_{hv0}$ is an indicator variable that takes value 1 if for household h in village v cultivated any of the promoted crops in their village at baseline, and 0 if they did not (“novice cultivator”). We present estimates of Equation 2 in Table 4, pooling across the two years following the initial demonstration year.

Focusing on control villages, we find that veteran cultivators are 255% more likely to cultivate promoted crops; this result corroborates the notion that veteran cultivators are persistently different from novice cultivators in their propensity to cultivate the promoted crops. Veterans in control villages are also proportionately more likely to cultivate the new seeds, underscoring the link between decisions to cultivate promoted crops and adopt the new seeds. In addition, veteran cultivators are 51% less likely to cultivate dry season paddy.

Allowing for the impact of demonstration to vary across veterans and novices does not reveal any significant heterogeneity in impacts on adoption of the new seeds, yield, fertilizer use, or profits (Table 4).

Table 4: Novices and veterans

	Promoted crops, One and two years post-demo					
	Improved seed	Cultivation	Yield ('000 BDT/ha)	Fertilizer ('000 BDT/ha)	Profit ('000 BDT/ha)	Paddy cultivation
	(1)	(2)	(3)	(4)	(5)	(6)
Veteran cultivator	0.054 (0.019) [0.006]	0.252 (0.059) [0.000]	-5.076 (6.825) [0.457]	0.061 (1.374) [0.965]	3.653 (2.437) [0.134]	-0.231 (0.096) [0.016]
Any demo in village	0.049 (0.011) [0.000]	0.056 (0.031) [0.076]	-5.090 (4.134) [0.219]	2.216 (1.495) [0.139]	-0.124 (0.687) [0.857]	-0.129 (0.118) [0.273]
Veteran cultivator × Any demo in village	-0.008 (0.024) [0.735]	-0.134 (0.062) [0.032]	10.748 (8.032) [0.181]	0.503 (2.143) [0.815]	-0.855 (2.666) [0.748]	0.118 (0.111) [0.285]
[Any demo in village] + [Veteran cultivator × Any demo in village]	0.041 (0.026) [0.120]	-0.078 (0.063) [0.218]	5.658 (7.539) [0.456]	2.718 (1.505) [0.076]	-0.979 (2.584) [0.706]	-0.011 (0.103) [0.917]
Control mean	0.020	0.099	44.983	5.038	1.632	0.454
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	4,422	4,422	729	729	4,422	2,253
# of clusters (Village)	70	70	63	63	70	70

Notes: Selected regression coefficients from Equation 2 are presented above, with selected sums of regression coefficients listed below, and included control variables also listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

However, similar impacts on adoption of the new seeds caused by demonstration operate differentially across novice and veteran cultivators of these crops. Among novices, demonstration increased cultivation of the promoted crop by 57% relative to the control group; this increase in cultivation among novices is fully joint with increased adoption of the new seeds. Moreover, the imprecisely estimated, large switch away from paddy as a result of demonstration observed in Table 3 is concentrated among novices, and the confidence interval does not allow us to reject that this decrease in paddy cultivation is fully accounted for by the increase in promoted crop cultivation (Columns 2 and 6, Table 4). While this estimate is also not statistically significant, it provides further suggestive evidence that demonstration enabled novice cultivators to cultivate pulses with new improved seeds, and in turn substitute away from paddy. In contrast, veteran cultivators do not respond to demonstration activities by altering their crop choice.

Taken together, these results suggest that returns to demonstration activities are homogeneous across novices and veterans: the promoted crops are traditionally grown across the study area, and novices and veterans alike know about the basic agronomy of these crops. Instead, demonstration principally triggers farmers to learn about the new seeds.¹⁶

¹⁶We revisit heterogeneity across novices and veterans in impacts of demonstration on adoption of the new seeds in Section 5.2.1, and present evidence consistent with the notion that novice cultivators were more uncertain about the productivity of the new seeds than veterans.

4.2 New improved seeds increase productive resilience

Empirical strategy Why would farmers choose to adopt the promoted varieties? The new seeds were developed to be resilient to flood-saline events which are relatively widespread across our study area (Appendix Figure A2), with an average of 25% of farmers reporting crop losses due to flooding in a given dry season.¹⁷ While we found no average impacts of demonstration on promoted crops’ yields or profits in Section 4.1, a simple treatment-on-the-treated back-of-envelope suggested that we may be under-powered to detect impacts given the levels of adoption triggered by demonstration. The possibility that the estimated null effects on yields and profits we report in Table 3 may mask impacts sufficiently large to drive adoption is further corroborated by the economically meaningful impacts of demonstration on input use that we observe.

To increase our power to detect any yield impacts of using saline flood-resilient seeds, we restrict our sample to cultivators of the promoted crops and compare yields of farmers who do and do not adopt the improved seeds. Of course, these farmers may differ. We instead make the following identifying assumption: absent the adoption of improved saline flood-resilient seed, the impact of exogenous saline-flood events on yields would have been the same across adopters and non-adopters.

To capture the yield impacts of using the improved seeds over our study period, we estimate:

$$Y_{hcv} = \alpha_{ct} + \beta_1 \text{Improved seed}_{hcv} + \beta_2 \text{Above median flood}_{hvt} + \beta_3 \text{Improved seed}_{hcv} \times \text{Above median flood}_{hvt} + \epsilon_{hcv} \quad (3)$$

where Y_{hcv} is household h ’s yield or profit per hectare for cultivated crop c in survey wave t . $\text{Improved seed}_{hcv}$ indicates that household h adopted improved seeds to cultivate crop c in survey wave t ; as the new flood saline-resilient varieties were not introduced until the demonstration year, we restrict our analysis to promoted crops during the demonstration year and one and two years post-demonstration. We use $\text{Above median flood}_{hvt}$ as our measure of flood-saline shocks; it is an indicator that takes a value 1 if village v ’s leave-household- h -out average flood exposure is above the median value in the data, and 0 otherwise. We plot variation in the modal value of above median flood exposure across villages in Appendix Figure A2, and we provide additional details on the construction of above median flood exposure in Appendix A. Given a flood realization, yields will be determined by crop-specific agronomic properties of the improved seed; hence, we control for crop-by-year fixed effects

¹⁷As noted above, while the incidence of tidal floods is relatively low during the dry season, their impact on soil salinity is strongest in this season.

α_{ct} .

Balance In order for our estimates of Equation 3 to capture the impacts of saline floods with and without the adoption of improved seed, it must be the case that, in a given year, saline flood events are exogenous and do not affect farmers’ decisions either to cultivate the promoted crops or to adopt the improved seeds. We present the following evidence consistent with these assumptions. First, in Appendix Table A20, we show that saline flood events do not affect farmers’ decisions to cultivate promoted crops, adopt improved seed (either unconditional or conditional on cultivating), or use fertilizer in that same year. We note that this exogeneity of flooding is consistent with both limited variation within Barisal in probability of flood exposure, and the unexpected nature of flooding in any given year preventing any anticipatory effects of floods on adoption. We present additional evidence of the former in Appendix Table A21: while above median flood exposure strongly predicts own flood exposure, it does not predict flood exposure in the subsequent year.

Results We present estimates of Equation 3 in Table 5; we find that adopting flood-saline-resilient seeds eliminates the adverse impacts of saline floods.

First, we establish large crop damage in the presence of saline floods: for farmers who do not cultivate improved seeds, yields and profits per hectare fall by 19% and 63%, respectively.

Second, these adverse impacts of floods go to zero for farmers cultivating improved seeds. The difference in impacts of saline floods for farmers who do and do not cultivate improved seeds is statistically significant for profits, and on the threshold of statistical significance for yields. This reduction in the impacts of saline floods on yields and profits corresponds to an increase in yields and profits per hectare of 8% and 45%, respectively.

The increase in productive resilience we document suggests that farmers may update more positively about the improved seeds in response to demonstration during a flood-saline shock. To test this hypothesis, we modify Equation 1 to allow for heterogeneity in the impacts of village assignment to demonstration with respect to the demonstration year flood-saline shock (Above median flood_{hv1}), and estimate:

$$\begin{aligned} \text{Improved seed}_{hcv} = & \alpha_{ct} + \beta_{1,t} \text{Above median flood}_{hv1} + \beta_{2,t} \text{Any demo in village}_v \\ & + \beta_{3,t} \text{Above median flood}_{hv1} \times \text{Any demo in village}_v + \epsilon_{hcv} \quad (4) \end{aligned}$$

We present estimates of Equation 4 in Table 6. We find suggestive evidence of stronger learning during flood-saline shocks: demonstration year flood-saline shocks increase adoption of improved seeds one year post-demonstration, and this result is driven entirely by villages

Table 5: Saline-resistant improved seed eliminates negative impacts of flood-saline shocks on profits

	Promoted crops, Cultivation = 1, Demo year and one and two years post-demo			
	Yield ('000 BDT/ha)		Profit ('000 BDT/ha)	
	(1)	(2)	(3)	(4)
Improved seed	2.497 (1.987) [0.209]	0.161 (2.165) [0.941]	2.363 (2.096) [0.260]	-0.399 (2.416) [0.869]
Above median flood		-7.383 (3.311) [0.026]		-6.111 (3.330) [0.067]
Improved seed $\times$ Above median flood		5.723 (3.558) [0.108]		6.874 (3.952) [0.082]
Control mean	36.49	39.78	7.72	9.71
Crop-wave FE	X	X	X	X
# of observations (Farmer-crop-wave)	1,336	1,336	1,336	1,336
# of clusters (Village)	90	90	90	90

Notes: Selected regression coefficients from Equation 3 are presented above, with included control variables listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

assigned to demonstration.¹⁸ In contrast, we find no effects of these flood-saline shocks on contemporaneous adoption (consistent with the exogeneity documented above) and no effects on adoption two years post-demo (as information generated by demonstration diffused through farmer groups).

While we presented evidence above that flood-saline shocks are exogenous, the decision to adopt the new seeds may be endogenous to other determinants of yields, which would prevent us from interpreting the increased flood-saline resilience under the new seeds as causal; we consider two possible forms of endogeneity, and we argue our estimate of the productivity impacts of the new seeds is likely conservative. First, the adoption of flood-saline-resilient seeds may be joint with other practices that help combat the negative effects of flood-saline shocks on yields; in this case, we interpret our estimates as the joint effect of these practices and the improved seeds. Second, it may be the case that farmers with higher or lower yields are more likely to adopt improved seeds; our estimates in Table 5 suggest that farmers who do and do not adopt improved seeds experience similar yields and profits

¹⁸We find similar impacts of demonstration year flood-saline shocks, available upon request, on fertilizer use, consistent with its association with the adoption of improved seed that we documented in Section 4.1.

Table 6: Demonstration year flood-saline shocks increase adoption of improved seed one year post-demonstration, but not contemporaneously or two years post-demonstration

	Promoted crops					
	Demo		One year		Two years	
	year		post-demo		post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Above median flood (Demo year)	0.041 (0.034) [0.217]	0.010 (0.019) [0.594]	0.035 (0.013) [0.007]	0.009 (0.015) [0.568]	-0.014 (0.025) [0.572]	0.004 (0.011) [0.703]
Any demo in village		0.127 (0.034) [0.000]		0.028 (0.014) [0.049]		0.054 (0.032) [0.091]
Above median flood (Demo year) × Any demo in village		0.014 (0.041) [0.731]		0.029 (0.023) [0.216]		-0.030 (0.034) [0.383]
[Above median flood (Demo year)] + [Above median flood (Demo year) × Any demo in village]		0.024 (0.037) [0.522]		0.038 (0.017) [0.031]		-0.026 (0.033) [0.438]
Control mean	0.098	0.000	0.035	0.000	0.053	0.000
Crop FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	1,642	1,642	3,289	3,289	2,244	2,244
# of clusters (Village)	96	96	96	96	70	70

Notes: Selected regression coefficients from Equation 4 are presented above, with included control variables listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

absent flood-saline shocks. This alleviates concerns that farmers may be positively selected on yields into adopting the improved seed, and suggest these estimates will not overstate the flood resilient impact of the improved seeds.

4.3 Demonstration persistently increases adoption, but these effects decrease over time

Empirical strategy We now study the dynamics of adoption in response to demonstration. We again estimate Equation 1, now allowing for the coefficient β_t on Any demo in village_{*v*} to vary across the two post-demonstration years.

Results Rather than an S-curve of technology adoption, estimating the impacts of any demonstration activity over the two dry seasons following the demonstration season reveals an inverted U-curve of adoption of the new seeds. First, any demonstration results in a stark

5.6 percentage point increase in adoption one year post-demonstration. This corresponds to a 155% increase off of 3.6% adoption in the control group, representing a 30 percentage point increase in adoption of the new seeds among cultivators of the promoted crops. Second, this effect decreases to 2.6 percentage points two years after the demonstration season. This long-term level of adoption is still high relative to the control group: it represents a 118% increase off of a 2.2% adoption in control villages, and conservatively implies a 14 percentage point increase in adoption rate among cultivators of the promoted crops.¹⁹

Table 7: Demonstration increases adoption of improved seed one and two years post-demonstration, while decentralizing demonstration only increases adoption one year post-demonstration

	Promoted crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.056 (0.017) [0.001]	0.028 (0.020) [0.167]	0.026 (0.012) [0.026]	0.022 (0.014) [0.106]	0.045 (0.013) [0.000]	0.027 (0.015) [0.068]
Shared demo in village		0.018 (0.022) [0.422]		0.019 (0.016) [0.236]		0.019 (0.017) [0.278]
Decentralized demo in village		0.065 (0.027) [0.018]		-0.006 (0.014) [0.658]		0.035 (0.018) [0.057]
Control mean	0.036	0.036	0.022	0.022	0.031	0.031
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	3,289	3,289	2,244	2,244	5,533	5,533
# of clusters (Village)	96	96	70	70	96	96

Notes: Selected regression coefficients from Equations 1 and 5 are presented above, with included control variables listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

These results suggest that adoption of the new seed varieties may be non-monotonic in learning, as impacts on adoption remain high overall but decrease over time. These dynamics are not driven by the control group catch-up, with control group adoption remaining stable and, if anything, falling over time, suggesting demonstration triggered the start of a learning

¹⁹In Appendix Table A10, we show that the decrease in the effect of any demonstration on adoption from one to two years post-demonstration is statistically significant, while we fail to reject that control group adoption does not change from one to two years post-demonstration.

process that had not yet begun in the control group two years post-demonstration.

4.4 Decentralizing demonstration increases adoption, but these additional impacts vanish over time

Empirical strategy We now leverage our multi-arm experimental design to estimate the additional impact of shared and decentralized demonstration relative to regular demonstration. Keeping similar notations as above, we now estimate:

$$\text{Improved seed}_{hcv} = \alpha_t + \beta_{1,t}\text{Any demo in village}_v + \beta_{2,t}\text{Shared demo in village}_v + \beta_{3,t}\text{Decentralized demo in village}_v + \epsilon_{hcv} \quad (5)$$

where $\text{Shared demo in village}_v$ is a binary variable indicating that demonstration implemented in village v followed a shared modality, while $\text{Decentralized demo in village}_v$ indicates that a decentralized modality was followed in village v . As we control for $\text{Any demo in village}_v$, $\beta_{2,t}$ and $\beta_{3,t}$ capture the additional impact of shared and decentralized demonstration in excess of the impact of regular demonstration (the omitted demonstration category), respectively. We estimate this model both allowing for the impacts of demonstration modalities to vary across post-demonstration years (Columns 2 and 4, Table 7), and pooling across post-demonstration years (Column 6, Table 7).

Balance and attrition In Table 2, we verify balance in key baseline household characteristics for shared and decentralized demonstration relative to regular demonstration. We fail to reject identical sample means across these three treatment arms, with one exception: number of social network connections is significantly lower among farmers in the shared demonstration arm. As social network connections is an important household characteristic with respect to learning, we present estimates of the impact of decentralizing demonstration with number of connections as a control in Appendix Table A17, with no qualitative effects on our results. We also note that 1 of 21 sample mean comparisons across our two balance tables being significant at the 5% level is consistent with expectation.²⁰

²⁰As in Section 4.1, in Appendix Table A24 we show that inference in our test of balance is not affected when we implement randomization inference, and we therefore apply robust inference clustered at the village-level. Relatedly, we present estimates of Equation 5 with randomization inference p-values included in Appendix Table A26, and patterns of statistical significance are unaffected. In Appendix Table A5, we establish that attrition one and two years post-demonstration was not significantly affected by decentralizing demonstration.

Results One year post-demonstration, when we found demonstration had the largest impacts on adoption, we find that decentralized demonstration doubles the impact of assigning demonstration on adoption of flood-saline-resilient seeds. Fully decentralized demonstration increases adoption by 6.5 percentage points in excess of a 2.8 percentage point effect of any demonstration, in total a 258% increase over 3.6% adoption in the control (Column 2, Table 7). In contrast, we fail to reject no additional impact of shared demonstration on adoption; however, the point estimate on shared demonstration corresponds to an economically significant increase in adoption of the new seeds (at 50% of control group adoption).

These additional effects of decentralizing demonstration vanish two years post-demonstration, leading the pooled impact of any demonstration to also fall, as documented in Section 4.3. The impact of regular demonstration is fairly stable over the two post-demonstration years (2.8 and 2.2 percentage points one and two years post-demonstration, respectively; Column 4, Table 7). However, the impact of decentralized demonstration sharply converges to the impact of regular demonstration, with its additional impact fading from 6.5 percentage points in the first year to an insignificant -0.1 percentage points in the second year.

4.5 Ruling out alternative mechanisms beyond learning

We consider three potential mechanisms besides learning that could explain the adoption dynamics we observe under demonstration and across demonstration modalities: seed recycling, seed supply, and forgetting.²¹ We discuss evidence in Appendix B that these mechanisms are unlikely to explain our results: seed was predominantly sourced from local markets, seed supply remained consistent even for control villages, and selection into adoption of the improved seeds became increasingly positive over time, respectively.

5 Model of learning and technology adoption

In the previous section we presented evidence of an “inverted U-curve” of adoption. Demonstration increased medium run adoption, and impacts persisted but decreased in the long run. If we believe that farmers continue to learn about the technology over time, these results suggest that adoption itself may not be monotonic in learning.

We therefore lean on theory to answer two questions: first, should we expect adoption to correspond directly to learning? and second, if not, can we infer learning from adoption?

²¹In our model in Section 5, we consider other mechanisms closely related to learning—non-Bayesian learning, and forward-looking experimentation one year post-demonstration—and argue these do not affect our conclusions.

First, in Section 5.1, we model demonstration as improving the precision of farmers’ signals of the returns to the demonstrated technology, in response to which farmers update their beliefs about these returns; our setup closely follows Besley and Case (1994).²² We show that adoption may be non-monotonic in learning when farmers adopt the demonstrated technology if they expect positive returns; initial learning introduces belief dispersion, which generates noisy adoption. We illustrate this phenomenon empirically in Section 5.2. In Section 5.2.1, we show noisy adoption is concentrated among novice cultivators, consistent with novice cultivators holding less certain prior beliefs which, in turn, magnifies belief dispersion (Conley and Udry, 2010). In Section 5.2.2, we show that demonstration caused profits to decrease and then increase, consistent with noisy adoption.

Second, in Section 5.3, we leverage the insight that adoption can still be used to infer the determinants of learning, at least during phases where adoption remains, on average, monotonic in learning. This insight generates tests of learning from scale, geography, and the set of demonstrators, and in turn explains the impacts of demonstration modalities on adoption.

5.1 Beliefs and technology adoption

Prior beliefs and the introduction of the new seeds Our model begins following the introduction of the new seeds by extension agents at the farmer group meetings. Extension agents can inform farmers about many aspects of the new seeds, such as their flood-saline-resilience, and farmers are already familiar with others, such as the suitability of the pulses to their plots and their household. However, as discussed in Section 2.3, farmers will remain uncertain about the productivity of the new seeds, A . Dissemination through meetings facilitates common initial beliefs across farmers about the seeds’ productivity A , as the meetings provide farmers the opportunity to discuss the new seeds with extension agents and each other in advance of any learning from demonstration (Chandrasekhar et al., 2020). We therefore begin with the simplifying assumption that all farmer group members hold a common normal prior about the returns to the new seeds.

$$A \sim \mathcal{N}(\mu_0, \sigma_0^2) \tag{6}$$

Signals and demonstration During the demonstration year, farmers observe an unbiased signal of the returns to the new seeds, Y . As in Besley and Case (1994) and Beaman et al.

²²In Appendix D, we present evidence that corroborates basic predictions from this model of learning: demonstration affects farmers’ beliefs that the new seeds are high yielding, which in turn is associated with adoption of the new seeds, and higher experienced profits during the demonstration year affect both beliefs about and adoption of the new seeds.

(2021), we model demonstration as affecting farmers’ technology adoption decisions through the variance of this signal, σ_Y^2 . This signal reflects farmers aggregating information about the productivity of the new seeds, gleaned from both their own demonstrations and the demonstrations of other farmers in their village.²³ As a consequence, the variance of the signal σ_Y^2 is affected by the demonstration modality assigned to the village, and may vary across farmers within village depending on who in the village demonstrates the new seeds.

$$Y|A = a \sim \mathcal{N}(a, \sigma_Y^2) \quad (7)$$

Updated beliefs and learning from demonstrations Following the demonstration year, and before the planting season one year post-demonstration, farmers update their beliefs about the returns to the new technology given their prior beliefs and the observed signal Y . We assume farmers update their beliefs following Bayes’ rule.²⁴

$$A|Y \sim \mathcal{N}(M, \sigma^2) \quad (8)$$

Farmers’ updated expected returns to the new technology, $M \equiv \ell Y + (1 - \ell)\mu_0$, are a weighted average of their prior expectations and the observed signal. The weight on the observed signal $\ell \equiv \sigma_0^2/(\sigma_0^2 + \sigma_Y^2)$ depends on the precision of the observed signal of the returns to the new seeds. We refer to ℓ as *learning*; consistent with this interpretation, farmers’ updated uncertainty, $\sigma^2 \equiv (1 - \ell)\sigma_0^2$, decreases with learning. When farmers do not observe any signal (“no learning”), that is $\sigma_Y^2 = \infty$, then learning $\ell = 0$. Alternatively, when farmers directly observe the returns to the new technology (“full learning”), that is $\sigma_Y^2 = 0$, then learning $\ell = 1$. As learning increases from no learning ($\ell = 0$) to full learning ($\ell = 1$), farmers observe an increasingly precise signal of the returns to the new technology and form increasingly precise updated beliefs.

Note that expected returns M vary across farmers, due to noise in the observed signal Y ; as we will see below, this variation is a key determinant of the impacts of learning on adoption of a new technology.

²³We assume farmers do not learn from other farmers’ decisions to demonstrate, as these decisions were made on the basis of common knowledge introduced at the farmer group meetings and therefore do not contain additional information about the new seeds.

²⁴There may be many reasons to expect deviations from Bayesian updating. Farmers may ignore that observed information may contain common sources, therefore overweighting positively correlated signals (Chandrasekhar et al., 2020). Alternatively, farmers may underweight signals from their social network (Conlon et al., 2022). In Appendix C, we consider deviations from Bayesian updating in the form of overreaction (underreaction) to the observed signal Y , and show they generate the same belief updating as under reduced (increased) prior certainty.

Updated beliefs and adoption one year post-demonstration Not all farmers optimally adopt the new seeds (Suri, 2011); farmers face known idiosyncratic costs of adopting the new seeds $C \sim F_C$, and adopt the new seeds if they expect positive net returns $A - C$.²⁵ Although we refer to C as “costs”, we interpret C as any known idiosyncratic factors that might drive decisions to adopt the new seeds, such as heterogeneous returns to cultivating pulses or to the flood-saline-resilience trait.²⁶ Letting D indicate that the farmer adopts the technology,

$$\begin{aligned} D &= \mathbf{1}\{\mathbf{E}[A|Y] - C > 0\} \\ &= \mathbf{1}\{M - C > 0\} \end{aligned} \tag{9}$$

Learning and adoption one year post-demonstration Our primary outcome of interest in Section 4, as is common in the literature on technology adoption, is the fraction of farmers who adopt the new seeds, or “population adoption” $\mathbf{E}[D|A = a]$. Population adoption corresponds directly to observed adoption in our experiment; it is the fraction of farmers who adopt the new seeds, conditional on a particular realization of the returns to the new seeds $A = a$. We emphasize the latter point: while farmers’ prior beliefs are over many potential realizations of the new seeds, it is one particular realization that is distributed to farmers in our experiment.

We make two additional comments on population adoption. First, population adoption depends on learning, as learning affects both farmers’ expectation of the returns to technology given their signal and also the precision of the signal. Second, population adoption is a nonlinear transformation of the farmer’s expected returns M , so the population average expected returns $\mathbf{E}[M|A = a]$ are insufficient to characterize population adoption.

Adoption of the new seeds depends on farmers’ expected returns and their costs; as the distribution of costs C is fixed, we first characterize the population distribution of expected

²⁵This model of adoption decisions implicitly imposes that farmers are myopic, that is they do not account for the impacts of their adoption on their future learning when deciding whether to adopt. These unaccounted “returns to experimentation” are largest when uncertainty about the technology is high, and therefore in general are decreasing in learning; in contrast, below we show that belief dispersion is increasing and then decreasing in learning, and therefore offers a parsimonious explanation of nonmonotonic adoption dynamics, the focus of our model.

²⁶This assumption rules out a model in which farmers learn both about average and idiosyncratic returns to the new seeds; while the predictions of this more general model are similar to ours, two empirical findings motivate our focus on learning the average returns to the new seeds. First, we fail to reject that farmers do not learn from geographic heterogeneity across plots in a test we develop in Section 5.3. Second, in Section 4.4 we presented evidence that long run learning converged across treatment arms, consistent with farmers learning the returns as information diffused; this is plausible for information about average returns, but not about idiosyncratic returns.

returns M in order to characterize population adoption.

$$M|A = a \sim \mathcal{N}(\ell a + (1 - \ell)\mu_0, \ell(1 - \ell)\sigma_0^2) \quad (10)$$

We interpret Equation 10 as follows. First, as learning increases from no learning to full learning, the population average expected returns $\mathbf{E}[M|A = a]$ shift monotonically from prior expectations μ_0 to the true returns a . Second, intermediate learning generates “belief dispersion”: some farmers observe relatively positive signals, while others observe relatively negative signals. Belief dispersion, that is the population variance of expected returns, is *nonmonotonic*, and specifically is increasing and then decreasing, in learning.

Proposition 1. *Population adoption is increasing in population average expected returns, and is increasing (decreasing) in belief dispersion if marginally adopting farmers have higher (lower) average expected returns than average farmers.*

Corollary 1. *Population adoption as a function of learning may be increasing, decreasing, increasing then decreasing, or decreasing then increasing.*

Proof. See Appendix C. □

5.2 Empirical evidence of noisy adoption

5.2.1 Stronger evidence of noisy adoption by novices than veterans

Key parameters underlying learning govern whether learning leads to increased adoption of a technology through an S-curve, with steadily growing adoption, or an inverted U-curve, with increasing and then decreasing adoption. We can see this in Equation 10. Population average expected returns will increase sharply with learning, generating the well-known “S-curve”, when true returns a are much larger than prior expectations μ_0 . On the other hand, belief dispersion will increase and then decrease sharply with learning, generating an “inverted U-curve”, when prior uncertainty σ_0^2 is high.

We apply these predictions to our finding that veteran cultivators did not experience larger impacts of demonstration on adoption of the new seeds than novice cultivators, although veteran cultivators were more likely to adopt the new seeds with or without demonstration; should we have expected to find heterogeneous impacts? These results pool across one and two years post-demonstration, and therefore pool across periods during which learning was low (one year post-demonstration) and high (two years post-demonstration). We hypothesize that novice cultivators have greater prior uncertainty than veteran cultivators: as novice cultivators have less experience with dissemination of improved seeds targeting

the promoted crops, we expect them to hold more uncertain prior beliefs. This additional prior uncertainty would lead to stronger inverted-U dynamics among novice cultivators, with relatively larger impacts one year post-demonstration than two years post-demonstration.

We test for stronger inverted-U dynamics among novice than veteran cultivators formally by re-estimating Equation 2 in Figure 5, which estimates heterogeneous impacts of demonstration with respect to veteran cultivator status, but with added interactions with two years post-demonstration.²⁷ The effect of demonstration on adoption decreases over time for novices, but not for veterans, and this difference is significant. Consequentially, the one year post-demonstration effect on adoption is driven primarily by novices, while the two year post-demonstration effect on adoption is driven primarily by veterans.

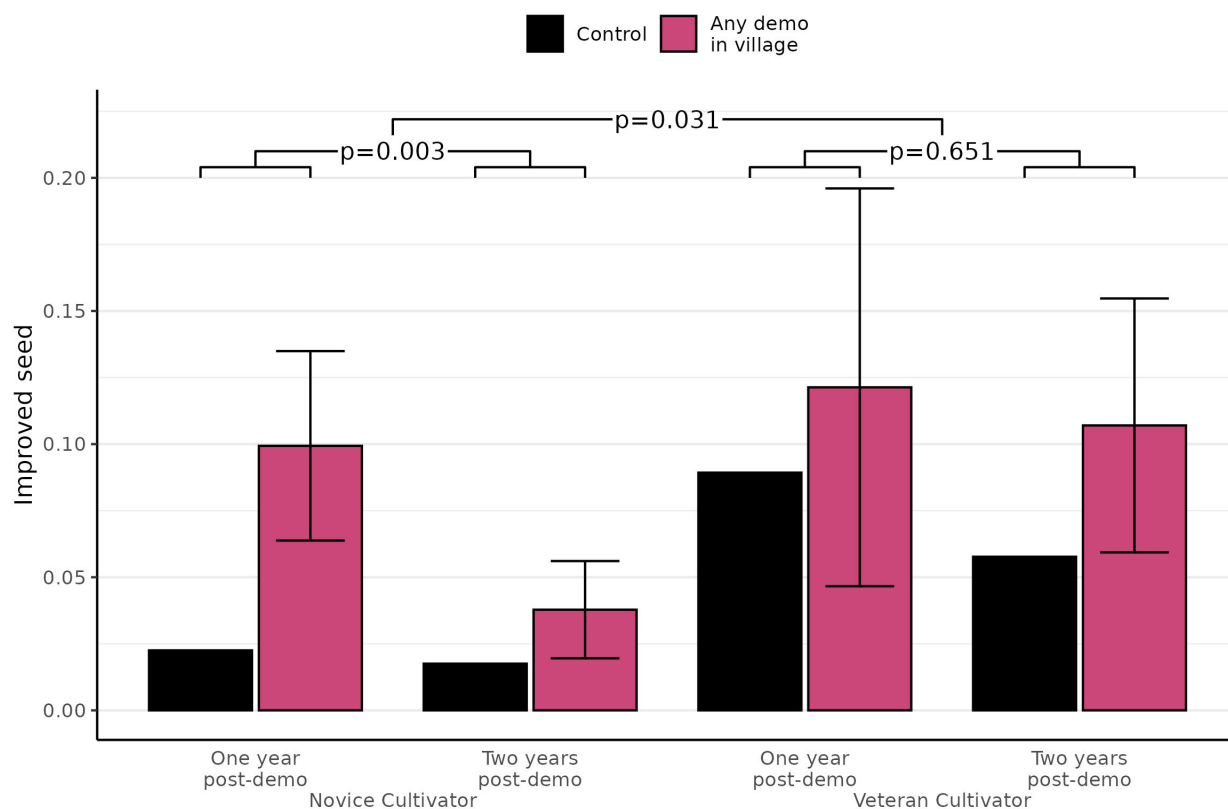


Figure 5: Heterogeneous adoption dynamics suggest higher returns to adoption and more precise prior beliefs for veterans

Notes: Control mean outcomes are presented in black, and treatment mean outcomes are presented in pink with 95% confidence intervals on treatment effects. Survey waves and veteran cultivator status are on the horizontal axis. p-values for tests of equality of treatment effects within cultivator status, and tests of equality of differences in treatment effects, are reported.

²⁷Coefficients are reported in Appendix Table A28.

5.2.2 Impacts on profits increase over time through positive selection into cultivation

While adoption mistakes imply that *initial* learning may not increase profits, *full* learning will necessarily maximize profits. That profits are eventually increasing in learning occurs due to improved selection on both C and A . As belief dispersion fades, farmers with higher returns to pulses or to the flood-saline-resilience trait (C) become relatively more likely to adopt. As average beliefs converge to the truth, technologies with higher returns (A) see larger increases in adoption.

We test for increased productivity due to improved selection into the new seeds over time by estimating dynamic effects of demonstration on yields and profits. We re-estimate Equation 1, which estimates the average effects of demonstration, but with added interactions with two years post-demonstration; a positive interaction coefficient is indicative of improving selection over time, while a positive effect of demonstration two years post-demo is indicative of long run learning increasing profits.

We find evidence of improving selection over time (Columns 1 and 3, Table 8), consistent with learning. Our average null results on yields and profits in Section 4.1 masked meaningful dynamics. Demonstration significantly *decreased* yields and profits one year post-demonstration, and impacts of demonstration on yields and profits significantly grew from one year to two years post-demonstration; long run learning appears to have undone adoption mistakes. Adding these coefficients, we find suggestive evidence that demonstration increased yields and profits two years post-demonstration. While our estimates are consistent with the impacts of the new seeds on productive resilience that we estimated in Section 4.2, a plausible benchmark for increased yields and profits under long run learning, we also lack the precision to rule out null effects.

Lastly, to separate different dimensions of selection, we additionally estimate specifications that control for farmer fixed effects; as yields are only observed when farmers cultivate, farmer fixed effects control for returns to pulses, isolating selection on returns to the flood-saline-resilient trait or on returns to the new seeds across space. We find that farmer fixed effects explain almost two thirds of the increase in impacts on yields (Column 2, Table 8). This suggests adoption mistakes are driven largely by farmers who would not otherwise (and should not) cultivate pulses.

5.3 Testing for learning from self and learning from others

We have established that technology adoption need not be monotonic in learning, even for Bayesian farmers; if so, what can technology adoption tell us about learning? We observe,

Table 8: Dynamic productive effects of demonstration suggest improved long run selection into improved seed

	Promoted crops, One and two years post-demo			
	Yield (’000 BDT/ha)		Profit (’000 BDT/ha)	
	(1)	(2)	(3)	(4)
Intercept	41.238 (2.216) [0.000]		1.335 (0.529) [0.012]	
Two years post-demo	5.143 (4.396) [0.242]	9.330 (3.954) [0.021]	1.529 (1.264) [0.227]	1.620 (1.449) [0.266]
Any demo in village	-8.241 (3.149) [0.009]		-1.659 (0.684) [0.015]	
Two years post-demo × Any demo in village	16.117 (5.470) [0.003]	5.972 (5.115) [0.246]	2.975 (1.379) [0.031]	2.594 (1.590) [0.106]
[Any demo in village] + [Two years post-demo × Any demo in village]	7.876 (5.858) [0.183]		1.316 (1.270) [0.303]	
Farmer FE		X		X
# of observations (Farmer-crop-wave)	912	912	5,533	5,533
# of clusters (Farmer group)	81	81	96	96

Notes: Selected regression coefficients from Equation 1, with included interactions with an indicator for two years post-demonstration, are presented above, with included control variables listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

building on Equation 10, that population technology adoption is a smooth function of learning. As a consequence, small exogenous increases in learning cause proportional changes in technology adoption. Relative impacts on adoption are therefore informative of relative impacts on learning. Our experimental results one year post-demonstration present an opportunity to estimate such relative impacts, revealing initial learning generated by demonstration. We use these impacts to propose tests for three key mechanisms underpinning learning proposed in Foster and Rosenzweig (1995): learning from self and learning from others, learning from scale, and learning from geographic heterogeneity.

We generate tests for each of these three mechanisms of learning by considering testable restrictions they imply for the variance of the signal observed by farmers, σ_Y^2 ; we summarize

predictions under each mechanism in Table 9. Across all three tests, and consistent with our conceptual framework in Section 3.2, we impose that farmers optimally aggregate signals across their own demonstration and demonstrations in their social network.²⁸

Table 9: Tests of determinants of learning

Learning from ...	Demo	# of connections to demo	Shared demo in village	Decentralized demo in village
self	+	·	·	·
others	·	+	·	·
scale	·	·	—	—
geographic heterogeneity	·	·	—	0

Notes: Predicted signs of coefficients in Equation 11, under different forms of learning, are presented above, with · when there is no prediction.

Learning from self and learning from others First, farmers may learn from themselves and from others. We suppose that farmers observe independent signals of the returns to the new seeds from each demonstration; however, farmers may observe a more precise signal of the returns to the new seeds from their own demonstration.²⁹ In this case, the impact of a farmer demonstrating on adoption, relative to the impact of a demonstration in that farmer’s social network, equals the relative strength of learning from self compared to learning from others.

Learning from scale Second, farmers may learn from the scale of demonstration. While farmers observe independent signals from each demonstration, the precision of each signal may grow with the size of the demonstration plot—farmers better observe the returns to the technology on larger demonstration plots.^{30,31}

²⁸While farmers may learn from demonstrations outside their social network, particularly once information has had sufficient time to diffuse beyond immediate connections, we build on a large literature on technology adoption in agriculture that provides evidence that learning from others is predominantly social and that social network connections provide a reasonable proxy for opportunities for social learning (e.g., [Conley and Udry, 2010](#); [BenYishay and Mobarak, 2019](#)).

²⁹We consider two possible sets of explanations of this difference in precision. The first is technological—if returns are highly idiosyncratic across plots or heterogeneous farming practices, then the returns under one farmer’s demonstration may not be informative for other farmers. The second is social—communication is inherently noisy and slow, and farmers may not immediately and fully learn from others’ demonstrations.

³⁰In [Foster and Rosenzweig \(1995\)](#), this growing precision is microfounded by each unit of area of a demonstration plot generating a signal, and imperfect correlation of these signals within demonstration plot.

³¹In an extreme, learning would depend only on the scale of demonstration, rather than the number of demonstration plots; our results in Section 4.4 strongly reject this.

We leverage the observation that moving from regular to shared to decentralized demonstration reduces the scale of demonstration, conditional on whether a farmer demonstrates and the number of demonstrations in the farmer’s social network. Under learning from scale, shared and decentralized demonstration therefore *decrease* learning conditional on whether a farmer demonstrates and the number of demonstrations in the farmer’s social network.

Learning from geographic heterogeneity Third, farmers may learn from geographic heterogeneity, that is observing the new seeds grown under different agronomic conditions. While each demonstration generates an independent signal of the returns to the new seeds under regular (as there is only one demonstration) or decentralized demonstration, all demonstrations occur under identical agronomic conditions under shared demonstration; shared demonstration only decentralizes socially, while decentralized demonstration equally decentralizes socially and geographically.

As a consequence, moving to shared demonstration, conditional on whether a farmer demonstrates and the number of demonstrations in the farmer’s social network, reduces the geographic heterogeneity under which demonstrations occur, and therefore *decreases* learning.

6 Learning from self and learning from others

In the previous section, we presented a model of learning and technology adoption that rationalized the “inverted U-curve” of adoption we observed, and concluded by deriving testable predictions (Table 9) of elements of the conceptual framework in Section 3.2 that motivated our experiment—the relative strength of learning from self and learning from others, learning from scale, and learning from geographic heterogeneity. In this section, we test those predictions—we find that just over half of learning is social, and fail to find evidence of learning from scale or learning from geographic heterogeneity.

Empirical strategy Our model-suggested tests depend on the effect of demonstrating and of connections to demonstrators, and the effects of decentralizing demonstration conditional on demonstrating and connections to demonstrators. Since farmers converge to similar levels of learning across experimental arms two years post-demonstration, we estimate these effects in the first post-demonstration year before information could diffuse beyond demonstrators’ networks; moreover, it is one year post-demonstration that we expect monotonic adoption with respect to learning at the initial learning generated by demonstration. In practice, we

estimate augmented versions of Equation 5:

$$\begin{aligned} \text{Any improved seed}_{hcv2} = & \alpha + \beta_1 \text{Any demo in village}_v \\ & + \beta_2 \text{Shared demo in village}_v + \beta_3 \text{Decentralized demo in village}_v \\ & + \gamma_1 \text{Demo}_{hcv} + \gamma_2 \# \text{ of connections to demo}_{hcv} + X'_{hcv} \delta + \epsilon_{hcv} \quad (11) \end{aligned}$$

Demo_{hcv} is a binary variable that indicates whether household h demonstrated the improved seed for promoted crop c during the demonstration season, while $\#$ of connections to demo_{hcv} captures the number of baseline social connections household h had to households who demonstrated the improved seeds for promoted crop c during the demonstration season.

We interpret the coefficients on Demo_{hcv} and $\#$ of connections to demo_{hcv} , γ_1 and γ_2 , as the effects of demonstrating and of connections to demonstrators on adoption of the improved seed. As the choice to demonstrate, and as a consequence connections to demonstrators, are not random, we present evidence below that these coefficients do reflect causal impacts; this includes adding controls X_{hcv} to Equation 11, for instance total number of connections, and placebo checks. Correspondingly, we interpret the coefficients on $\text{Shared demo in village}_v$, β_2 and β_3 , as the effects of decentralizing demonstration conditional on demonstration and connections to demonstrators.

We present estimates of Equation 11 in Column 1 of Table 10.

Learning from self and from others First, we find strong evidence of both learning from self and learning from others. Demonstrating increases the probability of adopting the new seeds by 12.7 percentage points, while one additional connection to a demonstrator increases the probability of adopting the new seeds by 1.7 percentage points; these effects are precisely estimated. As described in Section 5.3, the relative magnitudes of these effects approximates the relative strength of learning from self compared to learning from others; they imply learning from self is 7 times stronger than learning from an additional social connection. Scaling this average relative strength by demonstrators' average number of social connections (10.61) implies that 59% of learning is social.

Our estimates are comparable to existing estimates of the relative strength of learning from self and learning from others in the context of technology adoption in agriculture. Foster and Rosenzweig (1995) estimate the effect of others' experiences in one's village on profitability and initial adoption of improved seeds is between 1.2 and 2.7 times as large as the effect of own experience; these estimates imply between 54% and 73% of learning is social, qualitatively similar to our estimate of 59%. Beaman et al. (2021) estimate the effect of demonstration and number of connections to demonstrators on adoption of pit planting, and

Table 10: Impacts of decentralizing demonstration are fully explained by demonstration and connections to demonstrators

	One year post-demo				Two years post-demo
	Promoted crops			Placebo crops	Promoted crops
	Improved seed				
	(1)	(2)	(3)	(4)	(5)
Intercept	0.032 (0.014) [0.023]	0.029 (0.017) [0.093]		0.004 (0.002) [0.045]	0.023 (0.010) [0.024]
Any demo in village	0.025 (0.026) [0.334]	0.024 (0.025) [0.335]		0.031 (0.013) [0.015]	0.020 (0.014) [0.166]
Shared demo in village	0.000 (0.028) [0.994]	0.000 (0.028) [0.996]		-0.011 (0.016) [0.508]	0.012 (0.016) [0.455]
Decentralized demo in village	0.008 (0.031) [0.804]	0.008 (0.031) [0.792]		-0.016 (0.017) [0.330]	-0.018 (0.017) [0.285]
Demo	0.127 (0.032) [0.000]	0.128 (0.032) [0.000]	0.102 (0.043) [0.020]	0.017 (0.012) [0.152]	0.049 (0.018) [0.007]
# of connections to demo	0.017 (0.007) [0.013]	0.017 (0.007) [0.015]	0.026 (0.009) [0.006]	0.000 (0.003) [0.901]	0.001 (0.003) [0.753]
Veteran cultivator		0.029 (0.016) [0.079]			
# of connections		0.000 (0.002) [0.956]			
Farmer FE			X		
Pre-demo year sample only	X	X	X	X	X
# of observations (Farmer-crop-wave)	2,223	2,223	2,223	3,442	2,199
# of clusters (Village)	70	70	70	70	70

Notes: Regression coefficients from Equation 11 are presented above in Column 1, with variants on Equation 11 in Columns 2 through 5, and with included control variables listed below. Robust standard errors clustered at the village level are in parentheses, and p-values in brackets.

find demonstration increases adoption by between 12 and 19 times as much as connections to demonstrators; our estimate of 7 suggests social learning is somewhat stronger in our context, perhaps due to the social learning facilitated by farmer group participation, or the

greater importance of experiential learning for pit planting.

Differential selection into demonstration and social relationships Since household participation in demonstration was not randomly assigned, we perform four robustness checks to assess potential selection bias in our estimates of the impacts of demonstrating and of connections to demonstrators; we do not find evidence of meaningful selection bias.

- We compare demonstrators and non-demonstrators in Appendix Table A8. We fail to reject a joint test that they are observably similar, consistent with the relative homogeneity of farmer group members.³² However, we do observe statistically significant differences in veteran cultivator status and number of social connections, and we include these controls in Column 2 of Table 10.
- Multiple crops were demonstrated in many villages. We exploit within-household variation in demonstrating and connections to demonstrators by controlling for farmer fixed effects in Column 3 of Table 10, accounting for any potential differences in unobserved characteristics across demonstrators and non-demonstrators.
- We estimate precise zero effects of demonstrating and of connections to demonstrators on adoption of improved seed for placebo crops in Column 4 of Table 10.
- We estimate the effects of demonstrating and of connections to demonstrators on adoption of improved seed two years post-demonstration, at which point information generated by demonstration had diffused through the farmer groups. In Column 5 of Table 10, we find the effect of demonstrating fell by two thirds, and the effect of connections to demonstrators all but vanished; this is consistent with demonstrating and connections to demonstrators impacting adoption of improved seed through increased learning one year post-demonstration.

Learning from scale and geographic heterogeneity In contrast to the strong evidence we find for both learning from self and from others, we fail to find evidence of learning from scale or geographic heterogeneity in our context. In Column 1 of Table 10, we find no evidence

³²While it may be surprising that demonstrators and non-demonstrators are observably similar, we note that this masks heterogeneity across demonstration modalities (Appendix Table A9). The broad coverage of decentralized demonstration, and the requirement that demonstrators have adjacent plots for shared demonstration, limited scope for selection. As a consequence, regular demonstrators appear significantly positively selected relative to shared and decentralized demonstrators. The average similarity of demonstrators and non-demonstrators is therefore a consequence of the much larger number of demonstrators under shared and decentralized demonstration, who also drive our estimates of the effect of demonstrating and of connections to demonstrators.

that shared or decentralized demonstration affect adoption of the new seeds, after accounting for the average effects of demonstrating and of connections to demonstrators. Interpreting these null effects through the lens of our model predictions (Table 9) leads us to fail to reject the absence of learning from scale and from geographic heterogeneity in our context, as they imply that decreasing the scale of demonstration and reducing the geographic heterogeneity under which demonstrations occur do not significantly reduce learning.

We note that learning from scale and geographic heterogeneity may vary across technologies and approaches to demonstration. Learning from scale may only be of consequence when demonstration plots are meaningfully smaller than the typical scale of cultivation; as noted in Section 3.1, even under decentralized demonstration the “minikits” were sufficient to cultivate a not atypically-sized plot. Learning from geographic heterogeneity, in turn, depends crucially on whether observing the new seeds under geographically heterogeneous conditions within-village is informative about their returns.

7 Conclusion

We present experimental evidence that decentralizing agricultural demonstration at no additional cost accelerates learning about new technologies. The traditional approach centralizes all demonstration resources under a single carefully-selected farmer and relies on information to diffuse across their community. We show that dividing up the same total demonstration resources to many interested farmers accelerates learning, in aggregate. As more farmers demonstrate, they gain from experiential learning and information spreads further through their near-neighbors. Model-suggested tests imply that the smaller demonstration plots under decentralization do not undermine learning. Our experimental results thus support the notion that direct experience generated by small-scale demonstration and close communication are most useful for technology diffusion.

Our finding that decentralizing demonstration accelerates learning is informative for recent empirical and theoretical discussions in the literature about the nature of social learning. In one class of network diffusion models, increasing the number of demonstrations is more effective at triggering learning than carefully choosing the right demonstrators (Akbarpour et al., 2023). In other classes of models, choosing the wrong demonstrators can instead lead to persistently lower levels of adoption (Beaman et al., 2021), especially when image concerns can slow learning (Chandrasekhar et al., 2022; Banerjee et al., 2023). Simply letting anyone interested gain first-hand experience with a new technology is easily scalable, and our experiment shows it can increase learning without requiring operationally-costly network targeting.

Finally, given heterogeneity in the returns to new flood-saline-resilient seeds across growing conditions, farmers will make adoption mistakes as they learn about the new technology. This produces an inverted U-curve of technology adoption (rather than the standard S-curve) as adoption mistakes increase and then decrease as farmers learn. More precise communication strategies from the agricultural extension service would reshape these adoption dynamics by reducing adoption mistakes on the learning path, and may be worthwhile complements to providing farmers with first-hand experience of new technologies. These results and their implications are amplified by global climate change and the increased uncertainty it generates, as both the challenge of and returns to developing and disseminating customized agricultural technologies grow larger and larger.

References

- Agness, Daniel, Travis Baseler, Sylvain Chassang, Pascaline Dupas, and Erik Snowberg**, “Valuing the time of the self-employed,” *Review of Economic Studies*, 2025.
- Akbarpour, Mohammad, Suraj Malladi, and Amin Saberi**, “Just a Few Seeds More: The Inflated Value of Network Data for Diffusion,” 2023.
- Banerjee, Abhijit, Emily Breza, Arun G Chandrasekhar, and Benjamin Golub**, “When Less Is More: Experimental Evidence on Information Delivery During India’s Demonetisation,” *The Review of Economic Studies*, 2023.
- Beaman, Lori, Ariel BenYishay, Jeremy Magruder, and Ahmed Mushfiq Mobarak**, “Can network theory-based targeting increase technology adoption?,” *American Economic Review*, 2021, 111 (6), 1918–43.
- BenYishay, Ariel and Ahmed Mushfiq Mobarak**, “Social learning and incentives for experimentation and communication,” *The Review of Economic Studies*, 2019, 86 (3), 976–1009.
- Besley, T. and A. Case**, “Diffusion as a Learning Process: Evidence from HYV Cotton,” Papers 174, Princeton, Woodrow Wilson School - Development Studies 1994.
- Cai, Jing, Alain De Janvry, and Elisabeth Sadoulet**, “Subsidy policies and insurance demand,” *American Economic Review*, 2020, 110 (8), 2422–2453.
- Carter, Michael, Rachid Laajaj, and Dean Yang**, “Subsidies and the african green revolution: Direct effects and social network spillovers of randomized input subsidies in mozambique,” *American Economic Journal: Applied Economics*, 2021, 13 (2), 206–229.

- Chandrasekhar, Arun G, Esther Duflo, Michael Kremer, João F. Pugliese, Jonathan Robinson, and Frank Schilbach**, “Blue Spoons: Sparking Communication About Appropriate Technology Use,” Working Paper 30423, National Bureau of Economic Research September 2022.
- , **Horacio Larreguy, and Juan Pablo Xandri**, “Testing models of social learning on networks: Evidence from two experiments,” *Econometrica*, 2020, *88* (1), 1–32.
- Conley, Timothy G and Christopher R Udry**, “Learning about a new technology: Pineapple in Ghana,” *The American Economic Review*, 2010, pp. 35–69.
- Conlon, John J, Malavika Mani, Gautam Rao, Matthew W Ridley, and Frank Schilbach**, “Not Learning from Others,” 2022.
- Dar, Manzoor H, Alain De Janvry, Kyle Emerick, David Raitzer, and Elisabeth Sadoulet**, “Flood-tolerant rice reduces yield variability and raises expected yield, differentially benefitting socially disadvantaged groups,” *Scientific reports*, 2013, *3* (1), 3315.
- , – , – , **Erin M Kelley, and Elisabeth Sadoulet**, “Casting a Wider Net: Sharing Information Beyond Social Networks,” 2023.
- Dasgupta, Susmita, Md. Moqbul Hossain, Mainul Huq, and David Wheeler**, “Climate change, groundwater salinization and road maintenance costs in coastal Bangladesh,” *Policy Research working paper*, 2014, *WPS 7147*.
- Emerick, Kyle, Alain de Janvry, Elisabeth Sadoulet, and Manzoor H Dar**, “Technological innovations, downside risk, and the modernization of agriculture,” *American Economic Review*, 2016, *106* (6), 1537–61.
- Foster, A. and M. Rosenzweig**, “Learning by doing and learning from others: Human capital and technical change in agriculture,” *Journal of political Economy*, 1995, pp. 1176–1209.
- Gollin, Douglas, Casper Worm Hansen, and Asger Mose Wingender**, “Two blades of grass: The impact of the green revolution,” *Journal of Political Economy*, 2021, *129* (8), 2344–2384.
- Griliches, Zvi**, “Hybrid corn: An exploration in the economics of technological change,” *Econometrica, Journal of the Econometric Society*, 1957, pp. 501–522.

- Jones, Maria, Florence Kondylis, John Loeser, and Jeremy Magruder**, “Factor market failures and the adoption of irrigation in rwanda,” *American Economic Review*, 2022, *112* (7), 2316–52.
- Kondylis, Florence, John Ashton Loeser, Mushfiq Mobarak, Maria Ruth Jones, and Daniel Stein**, “Learning from self and learning from others: Experimental evidence from Bangladesh,” *World Bank Policy Research Working Paper*, 2023, (10545).
- , **Valerie Mueller, and Jessica Zhu**, “Seeing is believing? Evidence from an extension network experiment,” *Journal of Development Economics*, 2017, *125*, 1–20.
- Laajaj, Rachid, Karen Macours, Cergele Masso, Moses Thuita, and Bernard Vanlauwe**, “Reconciling yield gains in agronomic trials with returns under African smallholder conditions,” *Scientific reports*, 2020, *10* (1), 14286.
- Lane, Gregory**, “Adapting to climate risk with guaranteed credit: Evidence from Bangladesh,” *Econometrica*, 2024, *92* (2), 355–386.
- Patel, Dev**, “Environmental Beliefs and Adaptation to Climate Change,” 2024.
- Suri, Tavneet**, “Selection and comparative advantage in technology adoption,” *Econometrica*, 2011, *79* (1), 159–209.
- **and Christopher Udry**, “Agricultural technology in africa,” *Journal of Economic Perspectives*, 2022, *36* (1), 33–56.
- World Bank**, “World Development Report 2008: Agriculture for Development,” 2007.
- Young, H Peyton**, “Innovation diffusion in heterogeneous populations: Contagion, social influence, and social learning,” *American economic review*, 2009, *99* (5), 1899–1924.

A Data appendix

In this section, we define key variables used in our analysis; as appropriate, we follow [Jones et al. \(2022\)](#) and many of our definitions are identical to theirs. All agricultural variables are constructed using dry season data for the household’s two most important dry season plots with respect to cultivation of pulses.

Household variables: All household variables are constructed from the baseline.

- *Age of HHH*: Age of the household head.
- *Any pulses cultivation*: Any cultivation of pulses.
- *Veteran cultivator*: Any cultivation of promoted crops.
- *Any pulses improved seed*: Any use of improved seed for pulses; set to 0 when no pulses cultivation.
- *Any irrigation*: Any use of irrigation for paddy or pulses; set to 0 when no paddy or pulses cultivation.
- *Total plot area*: Area cultivated in hectares for paddy and pulses; set to 0 when no paddy or pulses cultivation.

Household-crop-survey wave variables:

- *Cultivation*: Indicator for any cultivation.
- *Improved seed*: Indicator for any use of improved seed; set to 0 when no cultivation.
- *Improved seed (Recycled)*: Indicator for any use of recycled improved seed; set to 0 when no improved seed use.
- *Yield ('000 BDT/ha)*: Harvested production, valued at median crop-survey wave sale prices, divided by area cultivated; winsorized at the 99th percentile and set to missing when no cultivation.
- *Fertilizer ('000 BDT/ha)*: Expenditures on fertilizer, divided by area cultivated; winsorized at the 99th percentile and set to missing when no cultivation. Only observed one year post-demo and two years post-demo.
- *Profit ('000 BDT/ha)*: Yield minus expenditures on hired labor per hectare minus household labor per hectare valued at 60% of the median agricultural wage (0.6×236 BDT/day) minus fertilizer expenditures per hectare; winsorized at the 99th percentile and then set to 0 when no cultivation. The choice of 60% follows a recommendation from [Agness et al. \(2025\)](#).

- *Profit (Demo year) ('000 BDT/ha)*: Demonstration year profits; set to missing when no demonstration or no reported cultivation. Expenditures on fertilizer are not subtracted as fertilizer was provided as part of the demonstration kit.
- *Area cultivated*: Area cultivated; set to missing when no cultivation.
- *Beliefs (High yielding)*: Indicator for reporting that one of the IAPP-promoted varieties is high yielding. Only observed one year post-demo.

Household-survey wave variables:

- *Paddy cultivation*: Any paddy cultivation.
- *Above median flood*: Indicator that leave-out flood exposure is above median. To construct leave-out flood exposure, we first construct a household's self-reported flood exposure as the fraction of crops cultivated by the household for which production was affected by flooding. We construct a household's associated leave-out flood exposure as the fraction of household-by-cultivated crop observations in that household's village *excluding that household* for which production was affected by flooding.

Experimental variables:

- *Any demo in village*: Indicator that the household's village was assigned to either *regular* demonstration, *shared* demonstration, or *decentralized* demonstration arms.
- *Shared demo in village*: Indicator that the household's village was assigned to the *shared* demonstration arm.
- *Decentralized demo in village*: Indicator that the household's village was assigned to the *decentralized* demonstration arm.
- *Demo*: Household-crop demonstration indicator. In analysis of impacts on placebo crops, this variable is the fraction of promoted crops that the household demonstrated.

Social network variables: All social network variables are constructed from the baseline. A household i and household j in the same village are defined as connected if either household i with household j or household j with household i reported discussing farming techniques in the last 3 months.

- *# of connections*: Number of social network connections, defined at the household level.
- *# of connections to demo*: Number of social network connections to demonstrators defined at the household-crop level. In analysis of impacts on placebo crops, this variable is the average number of social connections to demonstrators (defined at the household-crop level) across promoted crops.

B Robustness of experimental results

Selection of promoted crops In Section 3.3, we noted that our analysis in Section 4 restricted to promoted crops for power; this would introduce bias if the selection of promoted crops was endogenous to treatment. In Appendix Tables A6 and A7, we show that promoted crops, and the number of promoted crops, are orthogonal to treatment assignment; between 1.61 and 1.88 crops on average were promoted across treatment arms. Additionally, in Appendix Tables A18 and A19, we estimate the impacts of treatment assignment with crop-wave fixed effects to control for the possibility that different treatment arms may have caused different promoted crops to be selected. As we found no evidence of this selection, including these fixed effects does not affect our results.

Placebo crop analysis In Appendix Tables A11 and A12, we estimate the impacts of treatment on placebo crops. Given the positive impacts we estimate in our main analysis, we would be particularly concerned if we observed negative “impacts” on placebo crops, as these would be suggestive of positive selection of promoted crops. Instead, we find some evidence of positive spillovers of demonstration onto adoption of improved seed for placebo crops (although of much smaller magnitudes than our main effects).

Pre-demo analysis Complementing our balance tests in Section 4, in Appendix Tables A13 and A14, we find no differences in baseline outcomes across treatment arms.

Seed recycling and supply We find no impacts of demonstration on the use of recycled seed (Appendix Table A22), as farmers predominantly source the improved seed from Union-level seed multipliers or markets. We find no significant dynamics of adoption in the control group (Appendix Table A10), suggesting consistent availability of the improved seed.

Forgetting We find that impacts of demonstration on adoption converged across demonstration modalities (Table 7), including conditional on veteran status (Appendix Table A27), profits experienced by demonstrators (Appendix Table A1), and demonstration year flood-saline shocks (Table 6). Moreover, we find evidence of increasingly positive selection into improved seed over time: the share of adoption by veteran cultivators (Section 5.2.1) and the impacts of demonstration on profits (Section 5.2.2) are higher two years relative to one year post-demonstration. These results indicate increasing learning and are at odds with a story in which farmers’ knowledge fades over time (Foster and Rosenzweig, 1995).

C Model appendix

Proof. Proposition 1 and Corollary 1. First, we apply the law of iterated expectations to population adoption $\mathbf{E}[D|A]$, conditioning D on expected returns M .

$$\begin{aligned}\mathbf{E}[D|A] &= \mathbf{E}[\mathbf{E}[D|A, M]|A] && \text{LIE} \\ &= \mathbf{E}[\mathbf{E}[D|M]|A] && D \perp A|M \quad (\text{Equation 9}) \\ &= \mathbf{E}[F_C(M)|A] && D = \mathbf{1}\{C < M\} \quad (\text{Equation 9})\end{aligned}$$

Intuitively, population adoption is the expected fraction of farmers with costs less than their expected returns.

Second, we express population adoption as a function of population average expected returns and the population variance of expected returns, which we denote μ_M and σ_M^2 . Noting that expected returns $M|A$ are normally distributed (Equation 10), we parametrize $M|A \equiv \mu_M + \sigma_M Z|A$, where $\mu_M \equiv \ell A + (1 - \ell)\mu_0$, $\sigma_M^2 \equiv \ell(1 - \ell)\sigma_0^2$, and $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable. We write

$$\begin{aligned}\mathbf{E}[D|A] &= \mathbf{E}[F_C(M)|A] \\ &= \mathbf{E}[F_C(\mu_M + \sigma_M Z)|A]\end{aligned}$$

Proposition 1 Third, we differentiate population adoption with respect to population average expected returns and the population variance of expected returns. We let f_C denote the density of costs; differentiating yields

$$\begin{aligned}\frac{d}{d\mu_M} \mathbf{E}[D|A] &= \frac{d}{d\mu_M} \mathbf{E}[F_C(\mu_M + \sigma_M Z)|A] \\ &= \mathbf{E} \left[\frac{d}{d\mu_M} F_C(\mu_M + \sigma_M Z)|A \right] \\ &= \mathbf{E}[f_C(\mu_M + \sigma_M Z)|A] \\ &= \mathbf{E}[f_C(M)|A]\end{aligned}$$

Population adoption is increasing in population average expected returns, as $\frac{d}{d\mu_M} \mathbf{E}[D|A] = \mathbf{E}[f_C(M)|A]$, and the density of costs $f_C(\cdot)$ is positive. The intuition underlying the sign of this derivative is clear—when farmers expect higher returns, they are more likely to adopt. That this derivative is the expectation of the density of costs evaluated at expected returns also has a clear intuition; it is the density of farmers for whom expected returns are equal to costs, that is the density of marginal adopters. An increase in population average expected

returns therefore increases adoption by the density of marginal adopters.

$$\begin{aligned}
\frac{d}{d\sigma_M} \mathbf{E}[D|A] &= \frac{d}{d\sigma_M} \mathbf{E}[F_C(\mu_M + \sigma_M Z)|A] \\
&= \mathbf{E} \left[\frac{d}{d\sigma_M} F_C(\mu_M + \sigma_M Z) | A \right] \\
&= \mathbf{E} [Z f_C(\mu_M + \sigma_M Z) | A] \\
&= \mathbf{E} \left[\frac{M - \mu_M}{\sigma_M} f_C(M) | A \right] \\
&= \frac{\mathbf{E} \left[\frac{M - \mu_M}{\sigma_M} f_C(M) | A \right]}{\mathbf{E}[f_C(M)|A]} \mathbf{E}[f_C(M)|A]
\end{aligned}$$

The derivative of population adoption with respect to the population variance of expected returns is equal to the product of two terms: the average normalized distance between costs and population average expected returns among marginal adopters, and the density of marginal adopters. The intuition is as follows—increased belief dispersion “pushes” the distribution of beliefs away from population average expected returns, so adoption increases (decreases) when marginal farmers have relatively high (relatively low) expected returns compared to population average expected returns. Population adoption is therefore increasing (decreasing) in the population variance of expected returns if marginal farmers have higher (lower) average expected returns than average farmers.

Corollary 1 Lastly, we differentiate population adoption with respect to ℓ , applying our expressions for population average expected returns μ_M and the population variance of expected returns σ_M^2 .

$$\begin{aligned}
\frac{d\mathbf{E}[D|A]}{d\ell} &= \frac{d\mu_M}{d\ell} \frac{d\mathbf{E}[D|A]}{d\mu_M} + \frac{d\sigma_M}{d\ell} \frac{d\mathbf{E}[D|A]}{d\sigma_M} \\
&= (A - \mu_0) \frac{d\mathbf{E}[D|A]}{d\mu_M} + \frac{\sigma_0^2}{\sigma_M} \left(\frac{1}{2} - \ell \right) \frac{d\mathbf{E}[D|A]}{d\sigma_M}
\end{aligned}$$

As learning ℓ increases from 0 to 1, population average expected returns increase linearly from prior expected returns μ_0 to the true returns A —this force leads to adoption increasing (decreasing) with respect to learning when true returns A are above (below) prior expected returns μ_0 .

As learning ℓ increases from 0 to 1, the population variance of expected returns increases and then decreases; when marginal farmers have relatively high (relatively low) expected

returns compared to population average expected returns, this force causes adoption to increase and then decrease (decrease and then increase) with respect to learning.

Whether the effects of learning ℓ on population average expected returns or on the population variance of expected returns will dominate will depend on the underlying parameters. As a consequence, any of the following are possible: increasing adoption in learning (“S-curve”), decreasing adoption in learning, increasing then decreasing adoption in learning (“inverted U-curve”), or decreasing then increasing adoption in learning (“U-curve”). $\square$

Overreaction (underreaction) to signals generates identical adoption to a less (more) precise prior We proceed in two steps. First, we show that the distribution of expected returns is identical in a model in which farmers form posterior expected returns as if the observed signal has variance n times lower than its true variance and in a model in which farmers’ prior variance is n times higher. Second, we note that population adoption depends on beliefs only through the distribution of expected returns; therefore population adoption is identical under overreaction (underreaction) and less precise (more precise) priors.

Let $\hat{\ell} \equiv \frac{\sigma_0^2}{\sigma_0^2 + \sigma_Y^2/n} = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma_Y^2}$ denote learning under the perceived signal variance, σ_Y^2/n . Note that learning under the perceived signal variance is equal to learning in a model without misperceptions, but where the prior variance equals $n\sigma_0^2$.

Let $\widehat{M} \equiv \hat{\ell}Y + (1 - \hat{\ell})\mu_0$ denote expected returns under misperception of the signal variance. Note that when $n > 1$, $\hat{\ell}$ places too much weight (relative to the variance minimizing weight) on the observed signal Y , and farmers’ beliefs overreact to signals. Calculating the expectation and the variance of this expression, conditional on the technology, yields

$$\widehat{M}|A \sim \mathcal{N}(\hat{\ell}A + (1 - \hat{\ell})\mu_0, \hat{\ell}(1 - \hat{\ell})n\sigma_0^2)$$

Expected returns $\widehat{M}$ have the same distribution as in a model where the prior variance equals $n\sigma_0^2$ and there are no misperceptions.

D Learning the returns to a new technology explains medium run and long run impacts of demonstration

Our model of learning and technology adoption in Section 5, closely following [Besley and Case \(1994\)](#), assumed the following causal chain: demonstrations generate signals of the profitability of the new seeds, farmers update their beliefs about the profitability of the new seeds in response to observed signals, and decide to adopt the new seeds on the basis of these beliefs. We test this causal chain: does demonstration affect beliefs? are beliefs associated

with adoption? do profits affect beliefs? and adoption?

Empirical strategy We implement these tests of the causal chain underlying our model using versions of the following empirical specification:

$$Y_{hcv} = \alpha_{ct} + \beta_{1,t}\text{Any demo in village}_v + \beta_{2,t}\text{Shared demo in village}_v + \beta_{3,t}\text{Decentralized demo in village}_v + \gamma_t W_{hcv} + \epsilon_{hcv} \quad (\text{A1})$$

We report estimates of Equation A1 in Appendix Table A1, which we use to implement each of these four tests: demonstration affects beliefs (Column 1), beliefs are associated with adoption (Columns 2 and 3), profits affect beliefs (Column 4), and profits affect adoption (Columns 5 and 6).

Equation A1 extends Equation 5, used to estimate the impacts of demonstration and decentralizing demonstration on adoption, in three ways. First, it includes an additional covariate W_{hcv} : either household h 's belief that crop c is high yielding, measured one year post-demonstration (Columns 2 and 3), or household h 's demonstration year profits per hectare for crop c conditional on demonstrating (Columns 4, 5, and 6). Second, it varies the outcome Y_{hcv} : either beliefs (Columns 1 and 4), or adoption of the new seeds one year post demonstration (Columns 2 and 5) or two years post demonstration (Columns 3 and 6). Third, it includes additional controls: crop fixed effects in Columns 1, 2, and 3, to control for across-crop variation in beliefs that may correlate with propensity to adopt the new seed, and crop-by-treatment assignment fixed effects (which absorb controls for treatment assignment) in Columns 4, 5, and 6, to control for across-crop variation in profits generated by treatment assignment through the scale of demonstration.

Results We find that the way farmers learn in our data corroborates predictions from our model of learning (Appendix Table A1).

First, village assignment to demonstration positively affects farmers' beliefs that the promoted seeds are high-yielding one year post-demonstration (Column 1). Consistent with our results in Table 7, our point estimates imply that decentralized demonstration increases farmers' beliefs, albeit not statistically significantly.

Second, farmers' beliefs one year post-demonstration are strongly and precisely associated with increased adoption of the new seeds one year post demonstration (Column 2). However, this association between beliefs one year post-demonstration and adoption fades somewhat two years post-demonstration, in line with the notion that information about the new seeds

had diffused and farmers converged towards learning the true returns and an adoption steady state (Column 3).

Third, signals of the profitability of the new seeds (i.e., measured profits) generated by demonstration affect beliefs. Leveraging variation in profits conditional on crop and demonstration scale (i.e., crop-by-treatment assignment fixed effects), our point estimate implies a one standard deviation increase in demonstration year profits increases the probability the demonstrator believes the new seeds are high yielding by 7.4 percentage points, or a 26% increase (Column 4).

Lastly, signals of the profitability of the new seeds generated by demonstration affect adoption of the new seeds. A one standard deviation increase in demonstration year profits increases the probability the demonstration adopts the new seeds one year post-demonstration by 4.8 percentage points, 38% of the effect of demonstrating on adoption (Column 5). As with the association of adoption with beliefs, the effect of demonstration year profits on adoption fades two years post-demonstration (Column 6); farmers put less weight on information from their own demonstration as information about the new seeds diffuses.

E Appendix Tables

Table A1: Observed profits during demonstration shape beliefs, and in turn beliefs determine medium run, but not long run, adoption decisions

	Promoted crops					
	One year post-demo		Two years post-demo	One year post-demo		Two years post-demo
	Beliefs (High yielding)		Improved seed	Beliefs (High yielding)		Improved seed
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.084 (0.027) [0.002]	0.027 (0.020) [0.177]	0.017 (0.014) [0.228]			
Shared demo in village	-0.005 (0.029) [0.861]	-0.002 (0.022) [0.942]	0.019 (0.019) [0.317]			
Decentralized demo in village	0.044 (0.042) [0.297]	0.055 (0.027) [0.042]	-0.003 (0.015) [0.850]			
Beliefs (High yielding)		0.067 (0.021) [0.001]	0.024 (0.013) [0.061]			
Profit (Demo year) ('000 BDT/ha)				0.0026 (0.0012) [0.023]	0.0017 (0.0008) [0.038]	-0.0006 (0.0008) [0.461]
Demo only				X	X	X
Crop FE	X	X	X			
Crop-Treatment assignment FE				X	X	X
# of observations (Farmer-crop-wave)	2,934	2,934	2,037	170	184	123
# of clusters (Village)	91	91	67	55	59	42

Table A2: Agricultural descriptive statistics

	“Pulses”					
	Lentil	Mung	Mustard	Sesame	Wheat	Boro
	(1)	(2)	(3)	(4)	(5)	(6)
Yield ('000 BDT/ha)	36.89	39.36	39.74	26.91	46.34	88.09
Hired labor expenditures ('000 BDT/ha)	2.50	2.80	3.80	3.54	5.42	12.17
Household labor ('000 BDT/ha)	17.72	19.47	19.67	19.01	19.92	13.40
Fertilizer expenditures ('000 BDT/ha)	5.99	4.90	8.36	3.53	11.24	22.18
Seed expenditures ('000 BDT/ha)	3.97	2.62	1.57	1.71	2.90	3.66
Irrigation expenditures ('000 BDT/ha)	0.24	0.25	1.80	0.19	6.23	9.55
Profit ('000 BDT/ha)	6.48	9.32	4.54	-1.07	0.63	27.13
Any improved seed	0.106	0.399	0.306	0.218	0.601	0.599
Share irrigated	0.049	0.064	0.340	0.077	0.803	0.895
Area cultivated (ha)	0.118	0.161	0.109	0.135	0.101	0.297
Cultivated	0.165	0.236	0.046	0.025	0.065	0.388
# of observations (Farmer-crop-wave)	511	734	144	78	203	1,205

Notes: Sample averages of outcomes by crop across farmer-crop-survey wave observations are presented in this table. We restrict to observations one year and two years post-demo, in which input expenditures and household labor were collected disaggregated by crop. Profits in this table are calculated as yields net of expenditures on hired labor, fertilizer, seed, and irrigation, and household labor valued at 60% of the median wage of 236 BDT/day (choice of 60% following Agness et al., 2025). For reference, the median wage in our data is 236 BDT/person-day and the exchange rate was approximately 75 BDT = 1 USD over our study period.

Table A3: Sample descriptive statistics

	Sample								
	Pre-demo year	Demo year	One year post-demo	Two years post-demo	(3) + (4)	(2) + (3) + (4)	Pre-demo year sample		
							One year post-demo	Two years post-demo	(8) + (9)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Full sample	7,035 <1,407> {78}	4,750 <950> {109}	9,380 <1,876> {109}	6,150 <1,230> {77}	15,530 <3,106> {109}	20,280 <4,056> {109}	6,105 <1,221> {77}	6,015 <1,203> {77}	12,120 <2,424> {77}
Household analysis sample (Drop 2 Unions without demonstration)	6,895 <1,379> {76}	4,600 <920> {105}	9,075 <1,815> {105}	6,020 <1,204> {75}	15,095 <3,019> {105}	19,695 <3,939> {105}	5,985 <1,197> {75}	5,895 <1,179> {75}	11,880 <2,376> {75}
	2								
Promoted crops only	2,552 <1,308> {71}	1,642 <842> {96}	3,289 <1,676> {96}	2,244 <1,142> {70}	5,533 <2,818> {96}	7,175 <3,660> {96}	2,223 <1,133> {70}	2,199 <1,120> {70}	4,422 <2,253> {70}
		6(1,2)	6(3,4), 7(1,2),	6(5,6), 7(3,4),	3(1,2,5,6), 7(5,6), 8(3,4), A28(1)		10(1,2,3)	10(5)	4(1,2,5,6), A28(2,3)
Cultivated	164 <159> {39}	424 <371> {84}	610 <556> {76}	302 <282> {55}	912 <838> {81}	1,336 <1,209> {90}	433 <398> {58}	296 <277> {55}	729 <675> {63}
					3(3,4), 8(1,2)	5			4(3,4)
Placebo crops only	3,988 <1,308> {71}	2,568 <842> {96}	5,091 <1,676> {96}	3,466 <1,142> {70}	8,557 <2,818> {96}	11,125 <3,660> {96}	3,442 <1,133> {70}	3,401 <1,120> {70}	6,843 <2,253> {70}
							10(4)		

Notes: This table presents number of farmer-crop-wave observations, number of farmer-wave observations in angle brackets, and number of villages in curly brackets. Tables that use each sample for analysis are presented below these reported numbers of observations, with associated columns when not all columns use that sample.

Table A4: Attrition across control and treatment

	Mean (SD)		Coef. (SE) [p]
	Control	Any demo in village	Difference
	(1)	(2)	(2) - (1)
Surveyed (Demo year)	0.324 (0.469)	0.374 (0.484)	0.050 (0.030) [0.098]
Surveyed (One year post-demo)	0.840 (0.367)	0.875 (0.331)	0.035 (0.053) [0.503]
Surveyed (Two years post-demo)	0.811 (0.392)	0.866 (0.341)	0.055 (0.052) [0.292]
# of observations (Farmer)	281	1098	
# of clusters (Village)	15	61	
Omnibus F-stat [p]			1.22 [0.307]

Table A5: Attrition across treatment arms

	Mean (SD)			Coef. (SE) [p]	
	Regular demo in village	Shared demo in village	Decentral- ized demo in village	Difference	
	(1)	(2)	(3)	(2) - (1)	(3) - (1)
Surveyed (Demo year)	0.343 (0.475)	0.372 (0.484)	0.405 (0.492)	0.029 (0.025) [0.262]	0.062 (0.031) [0.042]
Surveyed (One year post-demo)	0.865 (0.342)	0.887 (0.316)	0.872 (0.335)	0.022 (0.036) [0.540]	0.007 (0.036) [0.848]
Surveyed (Two years post-demo)	0.850 (0.357)	0.885 (0.320)	0.861 (0.346)	0.034 (0.038) [0.369]	0.011 (0.035) [0.757]
# of observations (Farmer)	341	382	375		
# of clusters (Village)	19	21	21		
Omnibus F-stat [p]				1.40 [0.258]	1.95 [0.137]

Table A6: Promoted crop balance across control and treatment

	Mean		Coef. (SE) [p]
	Control	Any demo in village	Difference
	(1)	(2)	(2) - (1)
Wheat promoted	0.542	0.537	-0.005 (0.117) [0.965]
Lentil promoted	0.375	0.329	-0.046 (0.113) [0.686]
Mung promoted	0.375	0.390	0.015 (0.114) [0.894]
Sesame promoted	0.167	0.146	-0.020 (0.086) [0.814]
Mustard promoted	0.417	0.366	-0.051 (0.115) [0.659]
# of observations (Village)	24	82	
# of clusters (Village)	24	82	
Omnibus F-stat [p]			0.06 [0.998]

Table A7: Promoted crop balance across treatment arms

	Mean			Coef. (SE) [p]	
	Regular demo in village	Shared demo in village	Decentral- ized demo in village	Difference	
	(1)	(2)	(3)	(2) - (1)	(3) - (1)
Wheat promoted	0.593	0.571	0.444	-0.021 (0.135) [0.876]	-0.148 (0.137) [0.283]
Lentil promoted	0.333	0.250	0.407	-0.083 (0.124) [0.505]	0.074 (0.134) [0.581]
Mung promoted	0.333	0.393	0.444	0.060 (0.132) [0.653]	0.111 (0.134) [0.411]
Sesame promoted	0.222	0.071	0.148	-0.151 (0.095) [0.118]	-0.074 (0.107) [0.492]
Mustard promoted	0.370	0.321	0.407	-0.049 (0.131) [0.709]	0.037 (0.135) [0.785]
# of observations (Village)	27	28	27		
# of clusters (Village)	27	28	27		
Omnibus F-stat [p]				0.61 [0.690]	0.62 [0.682]

Table A8: Demonstrator characteristics

	Mean (SD)		Coef. (SE) [p]
	Non-demo	Demo	
	(1)	(2)	(2) - (1)
Age of HHH (Years)	48.50 (12.04)	48.59 (12.22)	0.09 (1.05) [0.931]
Any pulses cultivation	0.220 (0.415)	0.240 (0.428)	0.020 (0.032) [0.543]
Veteran cultivator (Any promoted crop cultivation)	0.112 (0.315)	0.147 (0.355)	0.035 (0.021) [0.087]
Any pulses improved seed	0.054 (0.226)	0.072 (0.258)	0.018 (0.020) [0.367]
Any irrigation	0.390 (0.488)	0.319 (0.467)	-0.071 (0.058) [0.223]
Total plot area (ha)	0.171 (0.286)	0.145 (0.231)	-0.026 (0.024) [0.285]
# of connections	9.06 (5.94)	10.61 (6.17)	1.54 (0.90) [0.088]
# of observations (Farmer)	762	279	
# of clusters (Village)	57	53	
Omnibus F-stat [p]			1.09 [0.379]

Table A9: Demonstrator characteristics across treatment arms

	Mean (SD)			Coef. (SE) [p]	
	Regular demo	Shared demo	Decentralized demo	Difference	
	(1)	(2)	(3)	(2) - (1)	(3) - (1)
Age of HHH (Years)	44.17 (11.28)	46.58 (11.77)	50.13 (12.33)	2.41 (2.45) [0.326]	5.97 (2.47) [0.017]
Any pulses cultivation	0.500 (0.511)	0.247 (0.434)	0.201 (0.402)	-0.253 (0.124) [0.042]	-0.299 (0.113) [0.009]
Veteran cultivator (Any promoted crop cultivation)	0.250 (0.442)	0.111 (0.316)	0.149 (0.358)	-0.139 (0.100) [0.166]	-0.101 (0.098) [0.303]
Any pulses improved seed	0.208 (0.415)	0.062 (0.242)	0.057 (0.233)	-0.147 (0.091) [0.110]	-0.151 (0.087) [0.084]
Any irrigation	0.250 (0.442)	0.321 (0.470)	0.328 (0.471)	0.071 (0.162) [0.662]	0.078 (0.150) [0.605]
Total plot area (ha)	0.147 (0.174)	0.179 (0.262)	0.129 (0.221)	0.032 (0.058) [0.588]	-0.018 (0.048) [0.712]
# of connections	12.58 (7.10)	7.60 (4.85)	11.73 (6.13)	-4.98 (2.22) [0.025]	-0.85 (2.40) [0.722]
# of observations (Farmer)	24	81	174		
# of clusters (Village)	13	19	21		
Omnibus F-stat [p]				3.06 [0.014]	2.75 [0.023]

Table A10: Dynamics of impacts of any demonstration are statistically significant, while falling adoption in control is not

	Promoted crops	
	One and two years post-demo	
	Improved seed	
	(1)	(2)
Intercept	0.036 (0.013) [0.005]	
Two years post-demo	-0.014 (0.013) [0.295]	-0.010 (0.013) [0.438]
Any demo in village	0.056 (0.017) [0.001]	
Two years post-demo $\times$ Any demo in village	-0.030 (0.018) [0.093]	-0.043 (0.019) [0.025]
Farmer FE		X
# of observations (Farmer-crop-wave)	5,533	5,533
# of clusters (Village)	96	96

Table A11: Limited impacts of demonstration on placebo crops

	Placebo crops, One and two years post-demo				
	Improved seed	Cultivation	Yield ('000 BDT/ha)	Fertilizer ('000 BDT/ha)	Profit ('000 BDT/ha)
	(1)	(2)	(3)	(4)	(5)
Any demo in village	0.014 (0.005) [0.005]	0.025 (0.018) [0.173]	0.862 (3.704) [0.816]	0.172 (1.058) [0.871]	0.334 (0.293) [0.253]
Control mean	0.006	0.057	35.851	5.371	0.371
Wave FE	X	X	X	X	X
# of observations (Farmer-crop-wave)	9,562	9,562	727	727	9,562
# of clusters (Village)	105	105	88	88	105

Table A12: No impacts of decentralizing demonstration on placebo crops

	Placebo crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.017 (0.004) [0.000]	0.023 (0.008) [0.004]	0.009 (0.008) [0.270]	0.011 (0.011) [0.314]	0.014 (0.005) [0.005]	0.018 (0.008) [0.016]
Shared demo in village		-0.006 (0.011) [0.591]		0.001 (0.012) [0.915]		-0.003 (0.010) [0.774]
Decentralized demo in village		-0.012 (0.009) [0.188]		-0.007 (0.009) [0.396]		-0.010 (0.008) [0.181]
Control mean	0.004	0.004	0.009	0.009	0.006	0.006
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	5,786	5,786	3,776	3,776	9,562	9,562
# of clusters (Village)	105	105	75	75	105	105

Table A13: Pre-demonstration outcomes are uncorrelated with demonstration

	Promoted crops, Pre-demo year					
	Improved seed	Culti- vation	Yield (^{'000} BDT/ha)	Fertilizer (^{'000} BDT/ha)	Profit (^{'000} BDT/ha)	Paddy cultivation
	(1)	(2)	(3)	(4)	(5)	(6)
	(0.014) [0.564]	(0.036) [0.911]	(11.107) [0.990]	(1.485) [0.893]	(0.876) [0.905]	(0.113) [0.544]
Any demo in village	-0.008 (0.014) [0.564]	-0.004 (0.036) [0.911]	0.143 (11.107) [0.990]	-0.199 (1.485) [0.893]	-0.104 (0.876) [0.905]	-0.069 (0.113) [0.544]
Control mean	0.027	0.067	41.175	4.470	2.090	0.457
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	2,552	2,552	164	164	2,552	1,308
# of clusters (Village)	71	71	39	39	71	71

Table A14: Pre-demonstration outcomes are uncorrelated with decentralizing demonstration

	Promoted crops, Pre-demo year	
	Improved seed	
	(1)	(2)
Any demo in village	-0.008 (0.014) [0.564]	-0.003 (0.016) [0.837]
Shared demo in village		-0.006 (0.011) [0.565]
Decentralized demo in village		-0.008 (0.010) [0.443]
Control mean	0.027	0.027
Wave FE	X	X
# of observations (Farmer-crop-wave)	2,552	2,552
# of clusters (Village)	71	71

Table A15: Impacts of demonstration are robust to restricting to pre-demonstration year sample

	Promoted crops, One and two years post-demo					
	Improved seed	Cultivation	Yield ('000 BDT/ha)	Fertilizer ('000 BDT/ha)	Profit ('000 BDT/ha)	Paddy cultivation
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.049 (0.013) [0.000]	0.040 (0.043) [0.359]	-1.936 (4.136) [0.640]	2.328 (1.080) [0.032]	-0.194 (0.838) [0.817]	-0.113 (0.113) [0.319]
Estimate with full sample	0.045	0.047	-3.401	2.019	-0.572	-0.111
Control mean	0.027	0.133	43.528	5.038	2.118	0.423
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	4,422	4,422	729	729	4,422	2,253
# of clusters (Village)	70	70	63	63	70	70

Table A16: Impacts of decentralizing demonstration are robust to restricting to pre-demonstration year sample

	Promoted crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.071 (0.020) [0.000]	0.038 (0.025) [0.129]	0.025 (0.012) [0.035]	0.022 (0.014) [0.116]	0.049 (0.013) [0.000]	0.031 (0.016) [0.061]
Shared demo in village		0.020 (0.028) [0.475]		0.017 (0.016) [0.290]		0.018 (0.019) [0.338]
Decentralized demo in village		0.071 (0.033) [0.031]		-0.006 (0.015) [0.675]		0.033 (0.019) [0.092]
Estimates with full sample	0.056	0.028 0.018 0.065	0.026	0.022 0.019 -0.006	0.045	0.027 0.019 0.035
Control mean	0.032	0.032	0.023	0.023	0.027	0.027
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	2,223	2,223	2,199	2,199	4,422	4,422
# of clusters (Village)	70	70	70	70	70	70

Table A17: Impacts of decentralizing demonstration are robust to controlling for number of connections

	Promoted crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.071 (0.019) [0.000]	0.035 (0.024) [0.147]	0.025 (0.012) [0.030]	0.021 (0.014) [0.139]	0.049 (0.013) [0.000]	0.028 (0.016) [0.072]
Shared demo in village		0.029 (0.027) [0.284]		0.021 (0.017) [0.229]		0.025 (0.020) [0.204]
Decentralized demo in village		0.071 (0.033) [0.029]		-0.006 (0.015) [0.695]		0.033 (0.020) [0.092]
# of connections	0.002 (0.002) [0.131]	0.002 (0.001) [0.118]	0.001 (0.001) [0.468]	0.001 (0.001) [0.242]	0.001 (0.001) [0.137]	0.002 (0.001) [0.094]
Estimates with full sample, no control	0.056	0.028 0.018 0.065	0.026	0.022 0.019 -0.006	0.045	0.027 0.019 0.035
Control mean	0.032	0.032	0.023	0.023	0.027	0.027
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	2,223	2,223	2,199	2,199	4,422	4,422
# of clusters (Village)	70	70	70	70	70	70

Table A18: Impacts of demonstration are robust to crop-wave fixed effects

	Promoted crops, One and two years post-demo					
	Improved seed	Culti- vation	Yield (‘000 BDT/ha)	Fertilizer (‘000 BDT/ha)	Profit (‘000 BDT/ha)	Paddy cultivation
	(1)	(2)	(3)	(4)	(5)	(6)
	(0.011) [0.000]	(0.027) [0.090]	(3.037) [0.143]	(0.690) [0.068]	(0.585) [0.295]	(0.095) [0.245]
Any demo in village	0.044	0.046	-4.448	1.258	-0.613	-0.111
Estimate with Wave FE	0.045	0.047	-3.401	2.019	-0.572	-0.111
Control mean	0.031	0.130	42.746	5.149	1.874	0.438
Crop-wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	5,533	5,533	912	912	5,533	2,818
# of clusters (Village)	96	96	81	81	96	96

Table A19: Impacts of decentralizing demonstration are robust to crop-wave fixed effects

	Promoted crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.055 (0.016) [0.001]	0.034 (0.017) [0.053]	0.027 (0.011) [0.019]	0.027 (0.014) [0.056]	0.044 (0.011) [0.000]	0.032 (0.012) [0.010]
Shared demo in village		0.003 (0.018) [0.863]		0.009 (0.017) [0.582]		0.006 (0.015) [0.691]
Decentralized demo in village		0.057 (0.025) [0.022]		-0.008 (0.015) [0.598]		0.029 (0.016) [0.076]
Estimates with Wave FE	0.056	0.028 0.018 0.065	0.026	0.022 0.019 -0.006	0.045	0.027 0.019 0.035
Control mean	0.036	0.036	0.022	0.022	0.031	0.031
Crop-wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	3,289	3,289	2,244	2,244	5,533	5,533
# of clusters (Village)	96	96	70	70	96	96

Table A20: Flood exogeneity

	Promoted crops, Demo year and one and two years post-demo			
	Cultivation	Improved seed	Improved seed Cultivation = 1	Fertilizer ('000 BDT/ha) Cultivation = 1
	(1)	(2)	(3)	(4)
Above median flood	0.014 (0.020) [0.486]	0.000 (0.010) [0.985]	-0.032 (0.049) [0.509]	0.366 (0.716) [0.610]
Control mean	0.171	0.071	0.418	6.563
Crop-wave FE	X	X	X	X
# of observations (Farmer-crop-wave)	7,161	7,161	1,336	1,336
# of clusters (Village)	96	96	90	90

Table A21: Flood first stage and serial correlation

	Promoted crops, Demo year and one and two years post-demo	
	Reported flood exposure	Above median flood
	(1)	(2)
Above median flood	0.051 (0.010) [0.000]	
Above median flood (1 year lag)		0.063 (0.044) [0.159]
Control mean	0.012	0.300
Crop-wave FE	X	X
# of observations (Farmer-crop-wave)	6,753	6,696
# of clusters (Village)	96	96

Table A22: Impacts of demonstration and decentralizing demonstration are not explained by seed recycling

	Promoted crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed (Recycled)					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.004 (0.009) [0.620]	-0.003 (0.009) [0.757]	0.004 (0.003) [0.256]	0.001 (0.003) [0.740]	0.004 (0.006) [0.483]	-0.001 (0.006) [0.836]
Shared demo in village		0.012 (0.011) [0.286]		0.005 (0.005) [0.320]		0.009 (0.008) [0.254]
Decentralized demo in village		0.009 (0.008) [0.242]		0.002 (0.004) [0.575]		0.007 (0.005) [0.209]
Control mean	0.013	0.013	0.002	0.002	0.010	0.010
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	3,289	3,289	2,244	2,244	5,533	5,533
# of clusters (Village)	96	96	70	70	96	96

Table A23: Balance with respect to any demonstration is robust to randomization inference

	Mean (SD)		Coef. (SE) [p]
	Control	Any demo in village	Difference
	(1)	(2)	(2) - (1)
Age of HHH (Years)	48.69 (11.93)	48.51 (12.10)	-0.18 (0.83) [0.832] {0.854}
Any pulses cultivation	0.203 (0.403)	0.215 (0.411)	0.012 (0.064) [0.850] {0.840}
Veteran cultivator (Any promoted crop cultivation)	0.117 (0.323)	0.115 (0.319)	-0.003 (0.061) [0.965] {0.959}
Any pulses improved seed	0.050 (0.218)	0.056 (0.229)	0.006 (0.027) [0.834] {0.814}
Any irrigation	0.463 (0.499)	0.398 (0.490)	-0.065 (0.111) [0.561] {0.552}
Total plot area (ha)	0.202 (0.300)	0.178 (0.282)	-0.024 (0.039) [0.537] {0.569}
# of connections	9.05 (6.65)	9.42 (5.97)	0.37 (1.62) [0.817] {0.815}
# of observations (Farmer)	281	1098	
# of clusters (Village)	15	61	
Omnibus F-stat [p]			0.15 [0.994] {0.998}

Table A24: Balance with respect to decentralizing demonstration is robust to randomization inference

	Mean (SD)			Coef. (SE) [p]	
	Regular demo in village	Shared demo in village	Decentral- ized demo in village	Difference	
	(1)	(2)	(3)	(2) - (1)	(3) - (1)
Age of HHH (Years)	48.71 (11.91)	48.32 (11.85)	48.53 (12.56)	-0.39 (0.97) [0.688] {0.706}	-0.18 (1.14) [0.872] {0.852}
Any IAPP cultivation	0.240 (0.428)	0.246 (0.431)	0.160 (0.367)	0.006 (0.068) [0.934] {0.930}	-0.080 (0.056) [0.153] {0.203}
Veteran cultivator (Any promoted crop cultivation)	0.100 (0.300)	0.126 (0.332)	0.117 (0.322)	0.026 (0.046) [0.573] {0.620}	0.018 (0.041) [0.667] {0.740}
Any IAPP improved seed	0.076 (0.266)	0.052 (0.223)	0.040 (0.196)	-0.024 (0.028) [0.392] {0.369}	-0.036 (0.027) [0.181] {0.179}
Any irrigation	0.367 (0.483)	0.406 (0.492)	0.419 (0.494)	0.039 (0.118) [0.740] {0.736}	0.052 (0.119) [0.661] {0.662}
Total plot area (ha)	0.163 (0.222)	0.198 (0.246)	0.171 (0.354)	0.035 (0.039) [0.369] {0.453}	0.008 (0.050) [0.873] {0.866}
# of connections	10.75 (6.43)	7.19 (5.04)	10.50 (5.76)	-3.56 (1.57) [0.024] {0.031}	-0.25 (1.75) [0.888] {0.886}
# of observations (Farmer)	341	382	375		
# of clusters (Village)	19	21	21		
Omnibus F-stat [p]				1.33 [0.263] {0.416}	0.96 [0.470] {0.630}

Table A25: Statistical precision of impacts of demonstration are robust to randomization inference, with exception of impacts on fertilizer use

	Promoted crops, One and two years post-demo					
	Improved seed	Cultivation	Yield ('000 BDT/ha)	Fertilizer ('000 BDT/ha)	Profit ('000 BDT/ha)	Paddy cultivation
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.045 (0.013) [0.000] {0.006}	0.047 (0.034) [0.163] {0.135}	-3.401 (3.344) [0.310] {0.462}	2.019 (0.884) [0.023] {0.127}	-0.572 (0.681) [0.401] {0.501}	-0.111 (0.095) [0.245] {0.231}
Control mean	0.031	0.130	42.746	5.149	1.874	0.438
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	5,533	5,533	912	912	5,533	2,818
# of clusters (Village)	96	96	81	81	96	96

Table A26: Statistical precision of impacts of decentralizing demonstration are robust to randomization inference

	Promoted crops					
	One year post-demo		Two years post-demo		One and two years post-demo	
	Improved seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Any demo in village	0.056 (0.017) [0.001] {0.014}	0.028 (0.020) [0.167] {0.337}	0.026 (0.012) [0.026] {0.064}	0.022 (0.014) [0.106] {0.190}	0.045 (0.013) [0.000] {0.007}	0.027 (0.015) [0.068] {0.198}
Shared demo in village		0.018 (0.022) [0.422] {0.522}		0.019 (0.016) [0.236] {0.246}		0.019 (0.017) [0.278] {0.346}
Decentralized demo in village		0.065 (0.027) [0.018] {0.016}		-0.006 (0.014) [0.658] {0.702}		0.035 (0.018) [0.057] {0.074}
Control mean	0.036	0.036	0.022	0.022	0.031	0.031
Wave FE	X	X	X	X	X	X
# of observations (Farmer-crop-wave)	3,289	3,289	2,244	2,244	5,533	5,533
# of clusters (Village)	96	96	70	70	96	96

Table A27: Conditional convergence across demonstration modalities

	Promoted crops			
	One year post-demo		Two years post-demo	
	Improved seed			
	(1)	(2)	(3)	(4)
Veteran cultivator	0.067 (0.029) [0.022]	0.067 (0.029) [0.022]	0.040 (0.014) [0.005]	0.040 (0.014) [0.005]
Any demo in village	0.077 (0.018) [0.000]	0.041 (0.024) [0.081]	0.020 (0.009) [0.030]	0.015 (0.011) [0.180]
Shared demo in village		0.026 (0.028) [0.354]		0.022 (0.012) [0.072]
Decentralized demo in village		0.073 (0.034) [0.029]		-0.004 (0.011) [0.721]
Veteran cultivator × Any demo in village	-0.045 (0.035) [0.195]	-0.028 (0.037) [0.449]	0.029 (0.022) [0.185]	0.043 (0.032) [0.189]
Veteran cultivator × Shared demo in village		-0.035 (0.038) [0.358]		-0.032 (0.044) [0.465]
Veteran cultivator × Decentralized demo in village		-0.012 (0.043) [0.777]		-0.009 (0.038) [0.815]
[Any demo in village] + [Veteran cultivator × Any demo in village]	0.032 (0.038) [0.403]	0.013 (0.044) [0.768]	0.049 (0.024) [0.047]	0.057 (0.035) [0.107]
[Shared demo in village] + [Veteran cultivator × Shared demo in village]		-0.009 (0.046) [0.842]		-0.010 (0.047) [0.828]
[Decentralized demo in village] + [Veteran cultivator × Decentralized demo in village]		0.061 (0.048) [0.208]		-0.013 (0.041) [0.755]
Control mean	0.023	0.023	0.018	0.018
Wave FE	X	X	X	X
# of observations (Farmer-crop-wave)	2,223	2,223	2,199	2,199
# of clusters (Village)	70	70	70	70

Table A28: Heterogeneous adoption dynamics suggest higher returns to adoption and more precise prior beliefs for veterans

	Promoted crops, One and two years post-demo		
	Improved seed		
	(1)	(2)	(3)
Intercept	0.036 (0.013) [0.005]	0.032 (0.014) [0.022]	0.023 (0.011) [0.047]
Veteran cultivator			0.067 (0.029) [0.022]
Two years post-demo	-0.014 (0.013) [0.295]	-0.009 (0.013) [0.483]	-0.005 (0.012) [0.673]
Veteran cultivator $\times$ Two years post-demo			-0.027 (0.026) [0.303]
Any demo in village	0.056 (0.017) [0.001]	0.071 (0.020) [0.000]	0.077 (0.018) [0.000]
Veteran cultivator $\times$ Any demo in village			-0.045 (0.035) [0.195]
Two years post-demo $\times$ Any demo in village	-0.030 (0.018) [0.093]	-0.046 (0.019) [0.015]	-0.057 (0.019) [0.003]
Veteran cultivator $\times$ Two years post-demo $\times$ Any demo in village			0.074 (0.034) [0.031]
Pre-demo year sample only		X	X
# of observations (Farmer-crop-wave)	5,533	4,422	4,422
# of clusters (Village)	96	70	70

F Appendix Figures

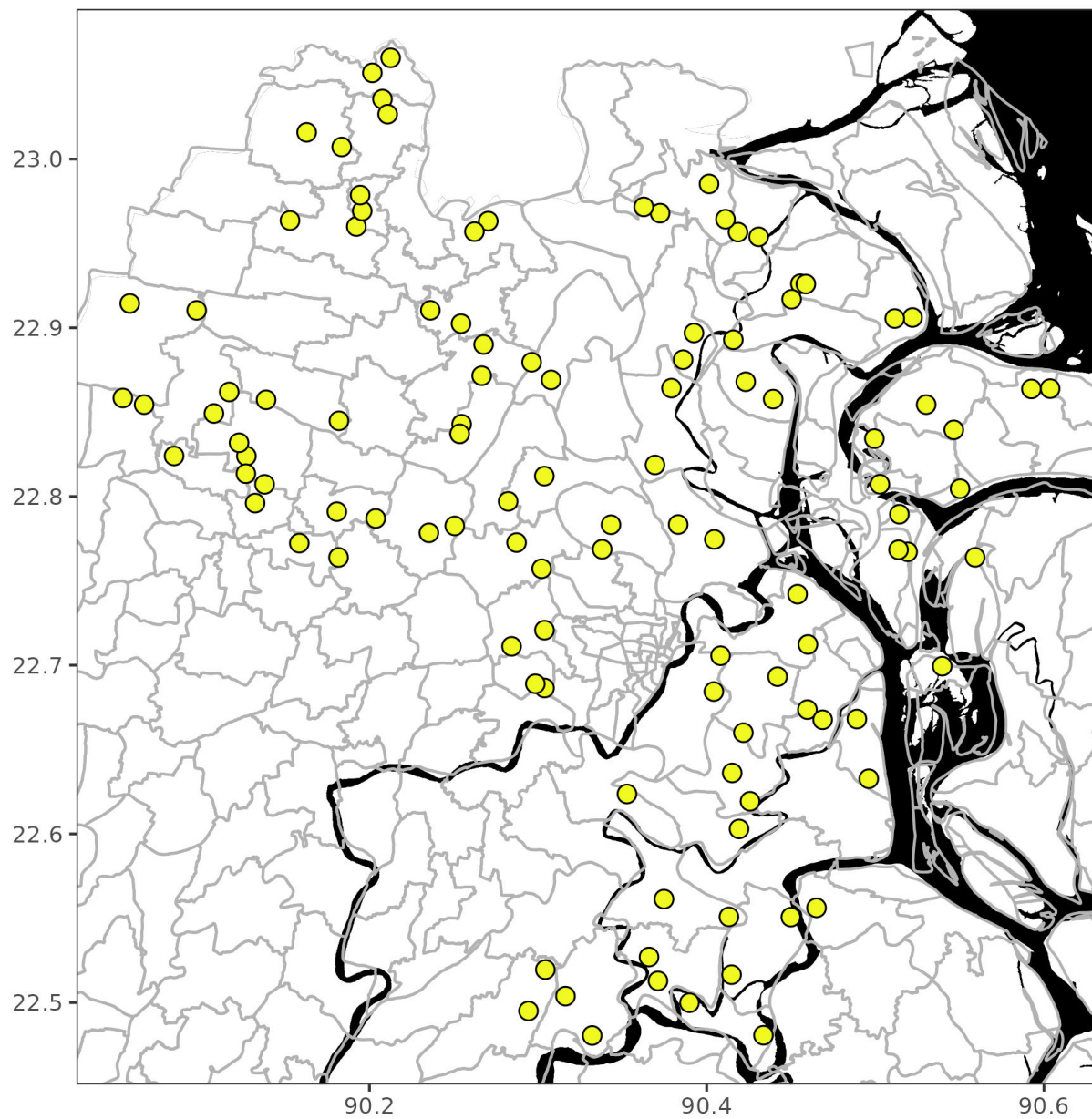
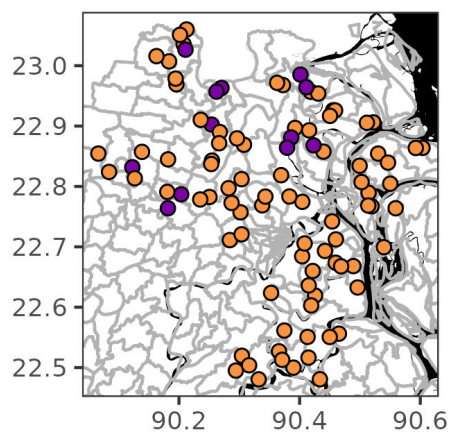


Figure A1: Village map

Notes: Our experimental sample of villages is plotted in **yellow**, with Union boundaries of Barisal in gray.

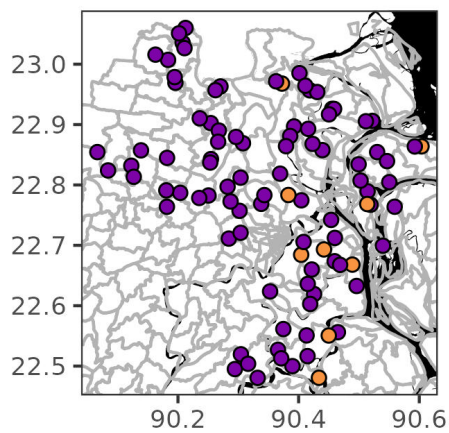
Demo year

Below median flood Above median flood



One year post-demo

Below median flood Above median flood



Two years post-demo

Below median flood Above median flood

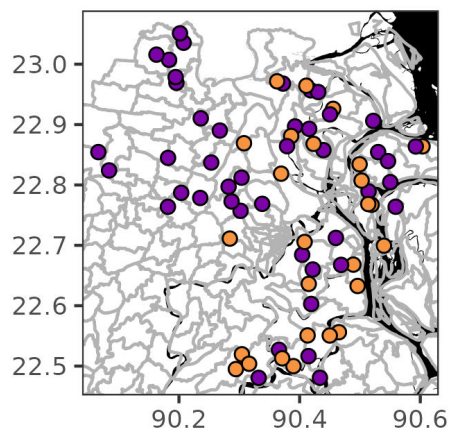


Figure A2: Floods